

46770 Integrated Energy Grids

Aleksander Grochowicz

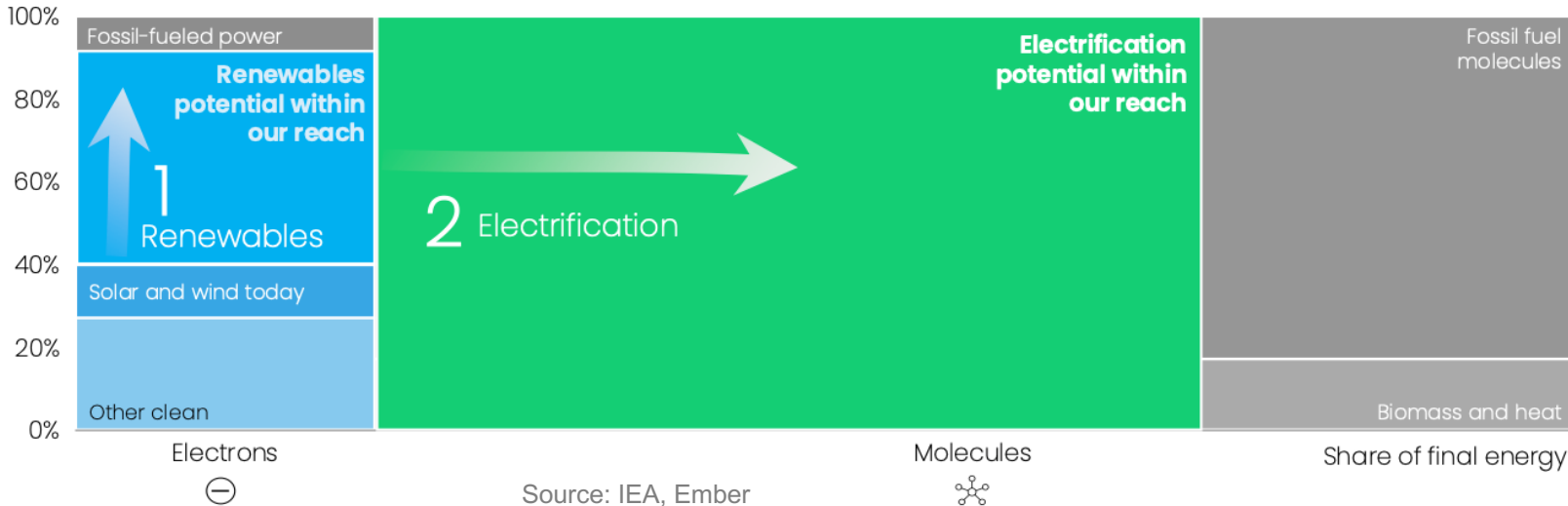
Lecture 11 – Multi-carrier energy systems I (heating and land transport)

Decarbonising the power sector is not enough...

The energy system comprises other sectors besides electricity, and they imply significant emissions

Global final energy demand in 2023

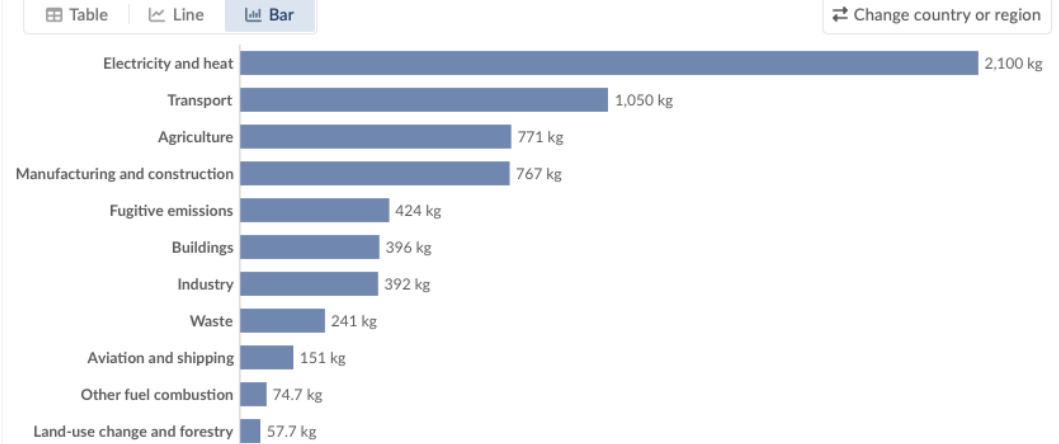
Share of final energy



Per capita greenhouse gas emissions by sector, World, 2023

Per capita greenhouse gas emissions are measured in kilograms of carbon dioxide-equivalents per person per year.

Our World in Data

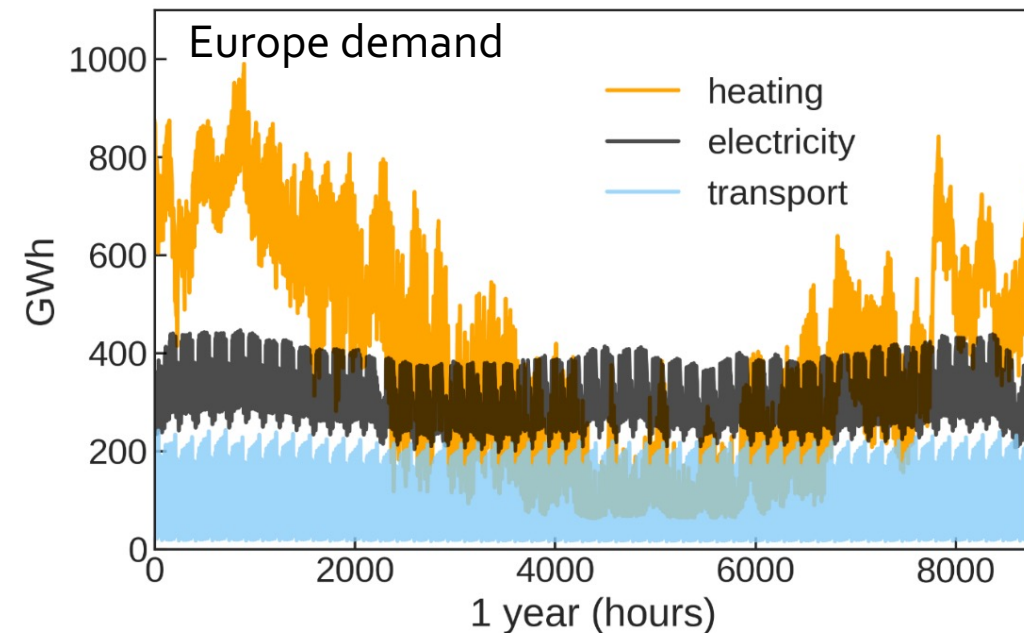


Learning goals

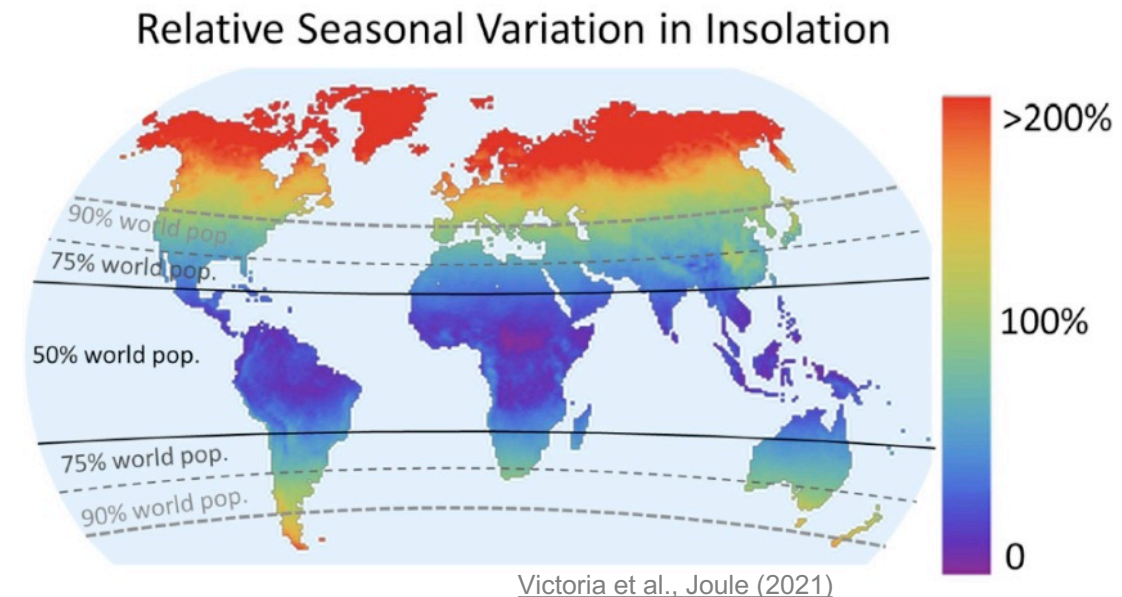
- Identify the main challenges and opportunities introduced by multi-carrier energy systems comprising power, heating and land transport.
- Describe modelling approaches that can be used to represent heating/cooling demand and heating generation and storage technologies.
- Describe modelling approaches that can be used to represent the electrification of land transport.
- Calculate representative time series for heating, cooling and land transport demand.
- Formulate optimisation problems comprising power, heating and land transport sectors in the computer.

Modelling heating/cooling demand and flexibility

Challenge: In Northern latitudes, heating demand shows much higher seasonal variation than electricity or transport demand.



Opportunity: Most of the world's population lives in the Sun Belt close to the equator where the solar resource is abundant and seasonal variation is low



Solar PV generation is synchronised with cooling demand !

Laine et al., Energy Environ. Sci., 2019

Zhu et al., Applied Energy, 2020.

DTU How can we model heating demand?

Conversely to electricity, we do not have historical data on heating demand with hourly resolution.

The profile of heating/cooling demand can be estimated using population-weighted ambient temperature.

Heating Degree Days or
Heating Degree Hours (HDH)

$$\text{HDH} = \begin{cases} T_{\text{threshold}} - T_{\text{amb}} & \text{if } T_{\text{amb}} \leq T_{\text{threshold}} \\ 0 & \text{if } T_{\text{amb}} > T_{\text{threshold}} \end{cases}$$

$$T_{\text{threshold}} \approx 17^{\circ}\text{C}$$

Cooling Degree Days or
Cooling Degree Hours (CDH)

$$\text{CDH} = \begin{cases} T_{\text{amb}} - T_{\text{threshold}} & \text{if } T_{\text{amb}} > T_{\text{threshold}} \\ 0 & \text{if } T_{\text{amb}} \leq T_{\text{threshold}} \end{cases}$$

$$T_{\text{threshold}} \approx 25^{\circ}\text{C}$$

The time series is then scaled based on the annual heating demand (for which we have historical data)

$$\begin{aligned} \text{heating demand}(t) &= \text{scaling factor} \cdot \text{HDH}(t) \\ &+ \text{hot water demand} \end{aligned}$$

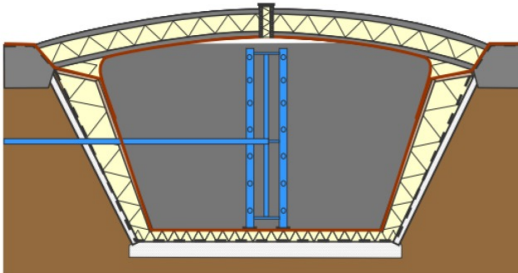
$$\text{cooling demand}(t) = \text{scaling factor} \cdot \text{CDH}(t)$$

Centralised systems

- Feasible for regions with high population density
- We need to model heat networks
- Technologies:



Combined Heat and Power



Central thermal energy storage
(self-discharge rate=0.02%/h)



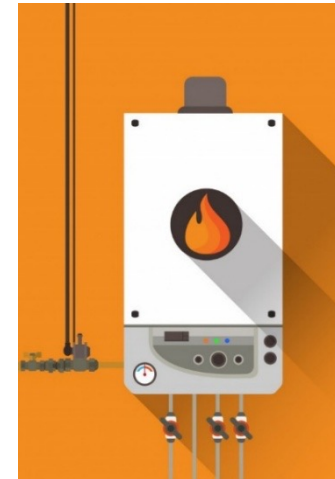
Heat pumps



Electric heating

Individual systems

- Typical in locations without district heating
- Technologies:



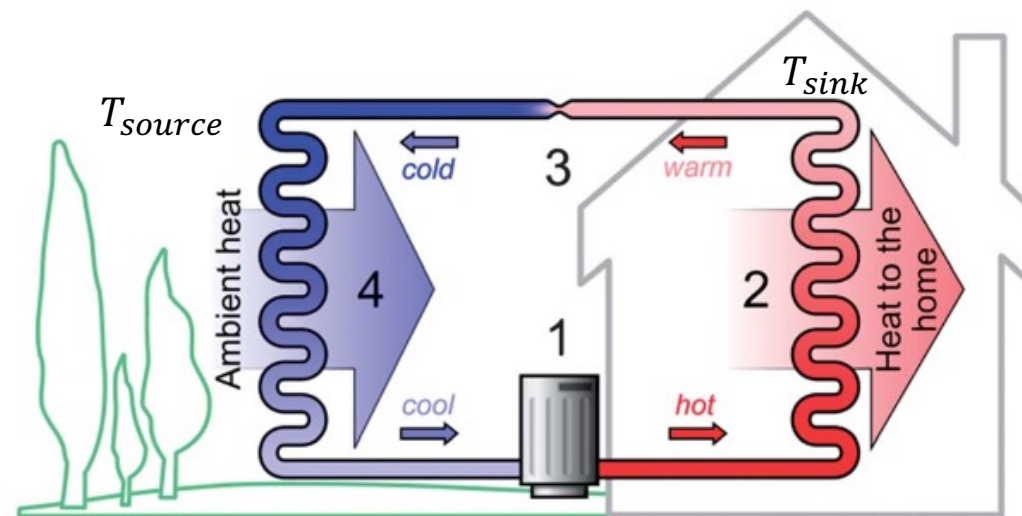
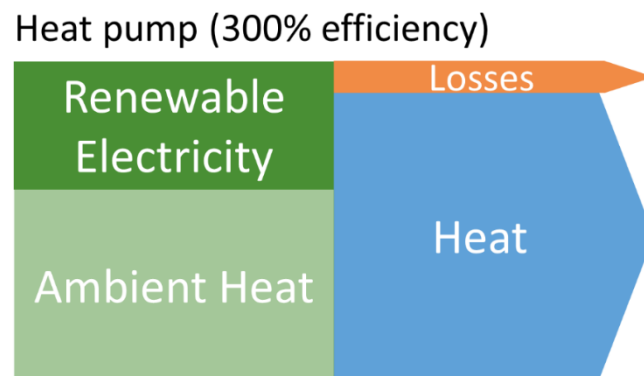
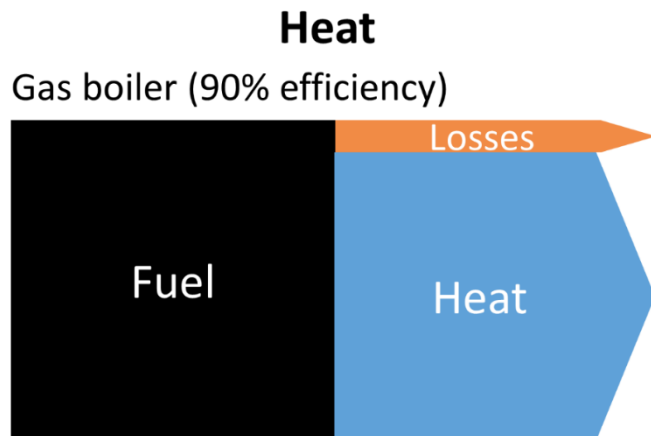
Gas/biomass
boilers



Individual thermal
energy storage
(self-discharge
rate=1.5%/h)

Heat pumps

Opportunities: Efficiency (Coefficient of Performance, COP) increases when using heat pumps, relative to fossil-based heating



Staffell, Energy Con. & Manag. 2012

But it is important to model that COP is lower when ambient temperature (T_{source}) is low

$$\text{COP}(\Delta T) = 6.81 - 0.121\Delta T + 0.00063\Delta T^2$$

$$\Delta T = T_{\text{sink}} - T_{\text{source}}$$

$$T_{\text{sink}} = 55^\circ\text{C}$$

We can also use ground-sourced or sea water-sourced heat pumps

Learnings from modelling the European energy systems

Optimisation problem

Most energy system models are just optimisation problems... with millions of variables and constraints.

We can start by calculating the optimal capacities for different technologies and how they need to be operated.

Recall our discussion on **joint capacity and dispatch optimisation**.

$$\min \left[\sum_{n,s} (\text{generation costs} + \text{storage costs} + \text{transmission costs}) + \sum_{n,s,t} \text{variable costs} \right]$$

subject to:

$$\sum_s g_{s,t,n} + \text{balance}_{t,n} = d_{t,n} \leftrightarrow \lambda_{t,n} \quad \forall t, n$$

$$\sum_{s,t} \text{CO}_2 \text{ emissions} \leq \text{CO}_2 \text{ limit} \leftrightarrow \mu_{\text{CO}_2}$$

$$g_{s,t} \leq \text{CF}_{s,t} \cdot G_s$$

$$G_{n,s} \leq G_{n,s}^{\max}$$

$$e_t = f(e_{t-1}, \text{charge}_t, \text{discharge}_t)$$

$$f_\ell \leq F_\ell$$

$$\sum_\ell K_{n\ell} f_\ell = p_n$$

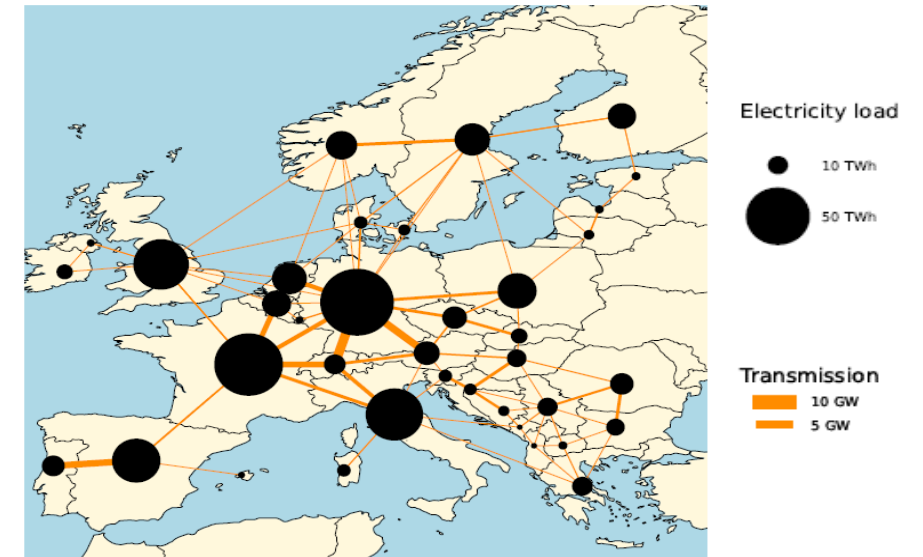
$$\sum_\ell C_{\ell c} x_\ell f_\ell = 0$$

n = nodes in the network

s = technologies

t = time steps

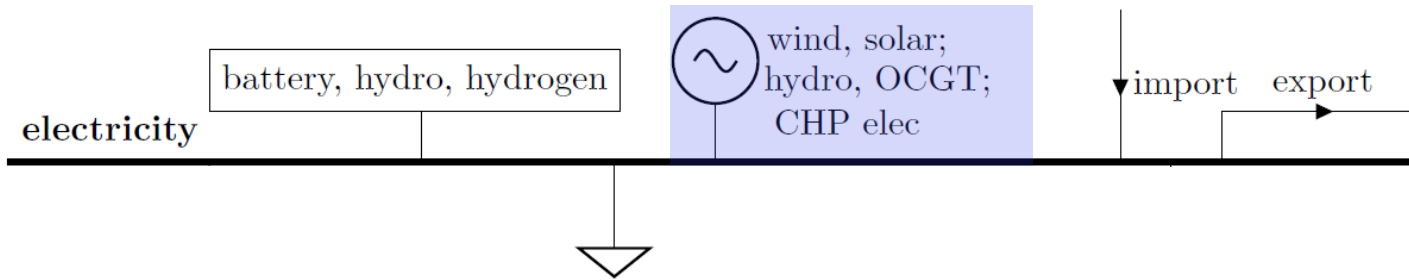
ℓ = links



* The model assumes long-term market equilibrium, perfect competition and foresight

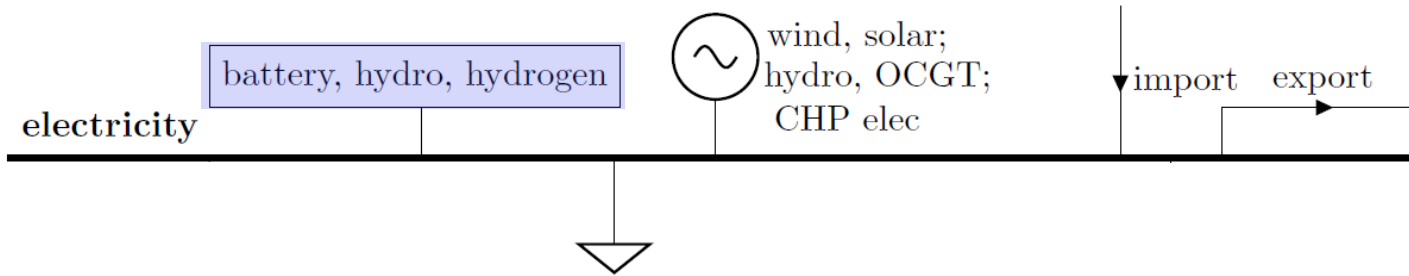
DTU Optimal power system (95% CO₂ reduction)

Let's start by obtaining insights from a simplified European power system.



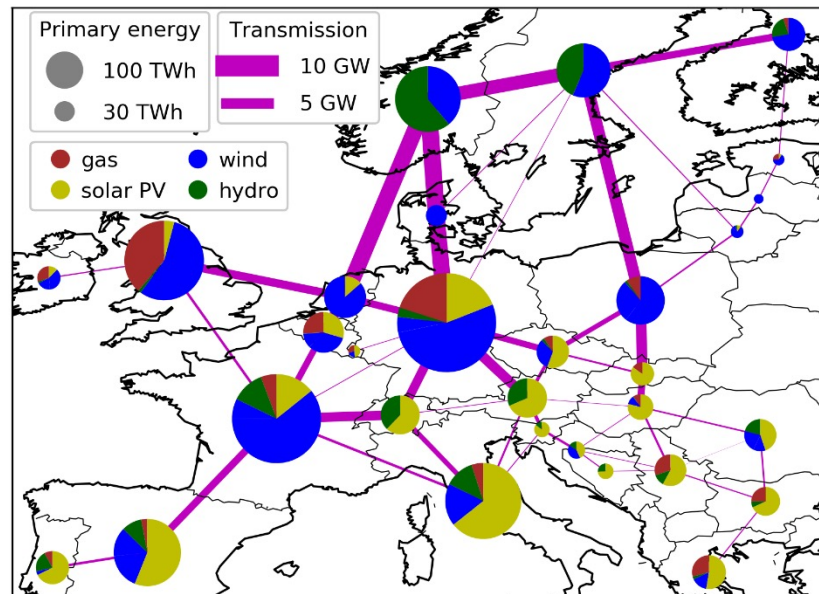
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Optimal power system (95% CO₂ reduction)

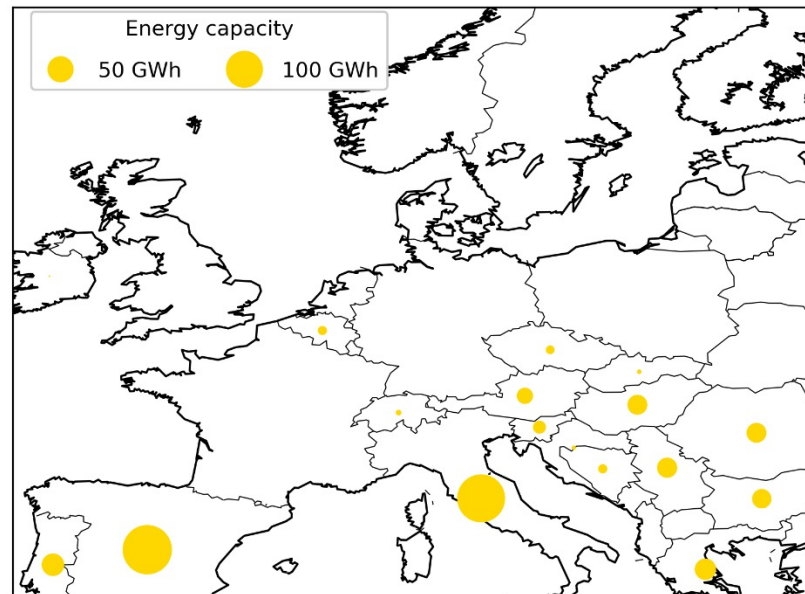
Primary energy



Wind generation represents in average 55% of the electricity demand, solar PV represents 33%

(570 GW onshore wind capacity, 60 GW offshore wind capacity)

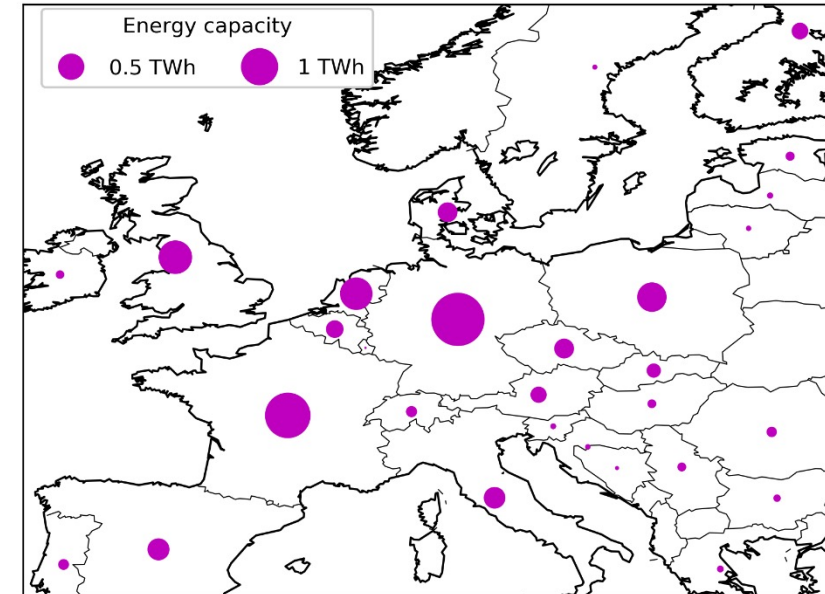
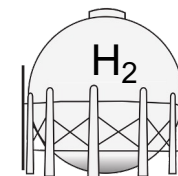
Electric Batteries



Southern countries: PV + batteries

$$\frac{\text{energy capacity}}{\text{power capacity}} \approx 6 \text{ hours}$$

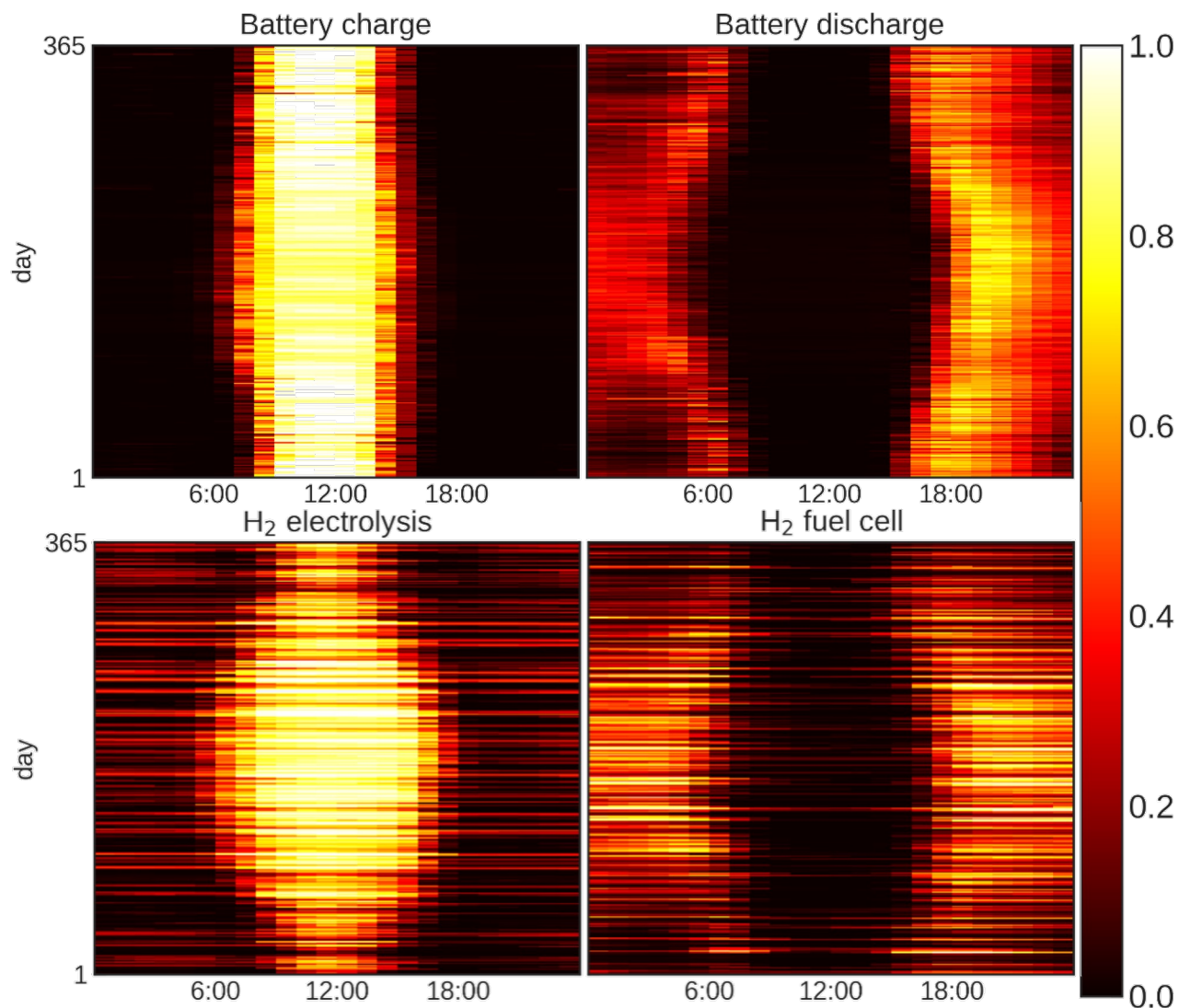
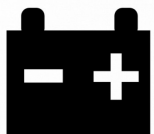
Hydrogen storage



Northern countries: wind + hydrogen storage + interconnections

$$\frac{\text{energy capacity}}{\text{power capacity}} \approx 2 \text{ days}$$

Victoria et al., Energy Conversion and Management (2019)



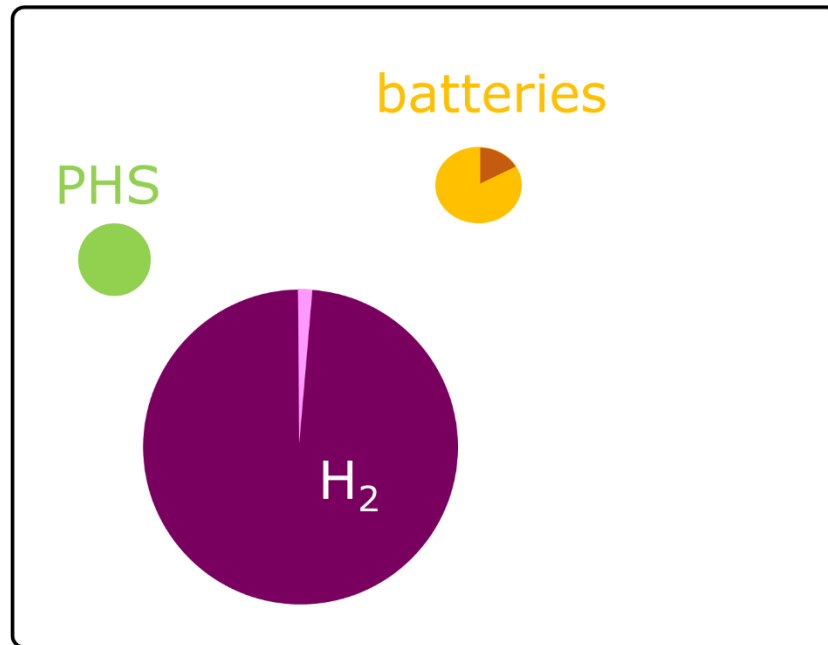
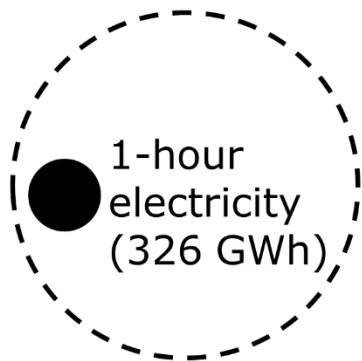
As expected, batteries charge during the day and discharge during the night.

Hydrogen storage operation is impacted by wind generation fluctuations with weekly frequency.

Victoria et al., Energy Conversion and Management (2019)

How much storage capacity is needed?

1-day electricity
load (7.8 TWh)



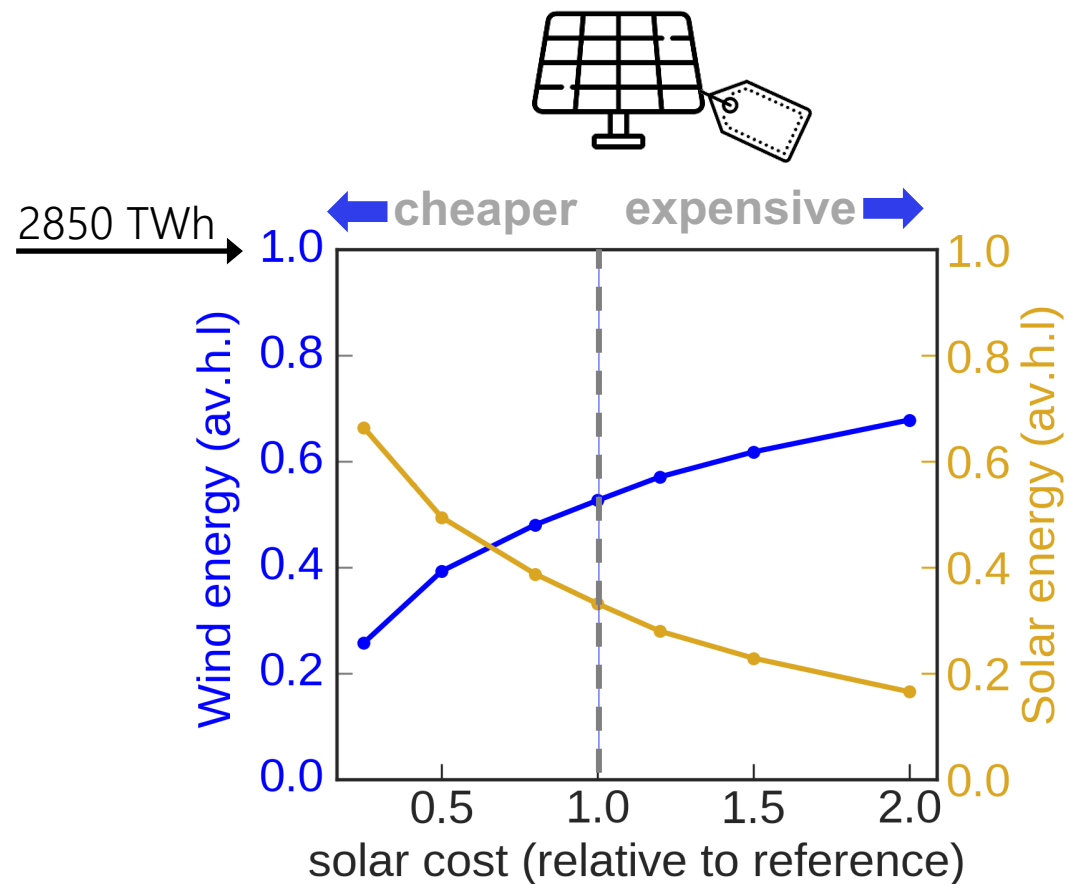
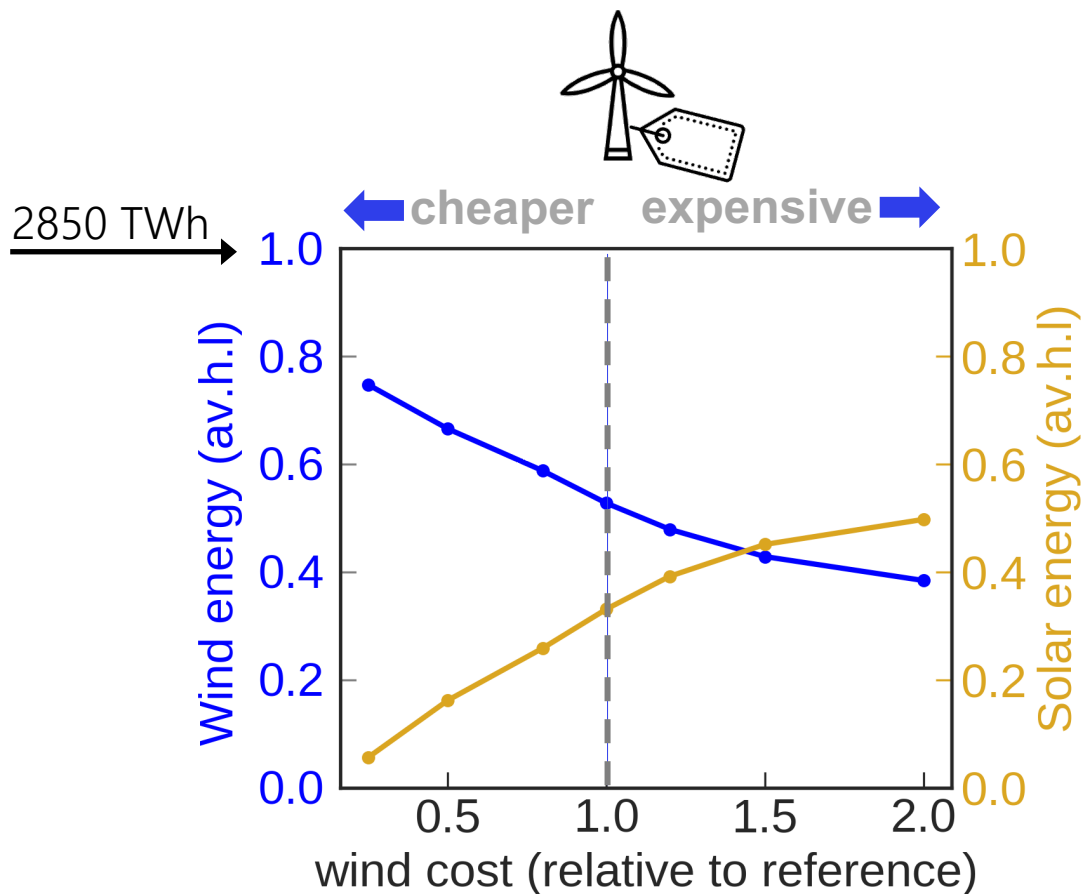
Victoria et al., Energy Conversion and Management (2019)

Optimal power system (95% CO₂ reduction)

- Pumped Hydro Storage ~ 0.9 average hourly electricity demand (exogenously fixed)
- Battery ~ 1.4 hourly electricity demand
- H₂ storage ~ 20 average hourly electricity demand

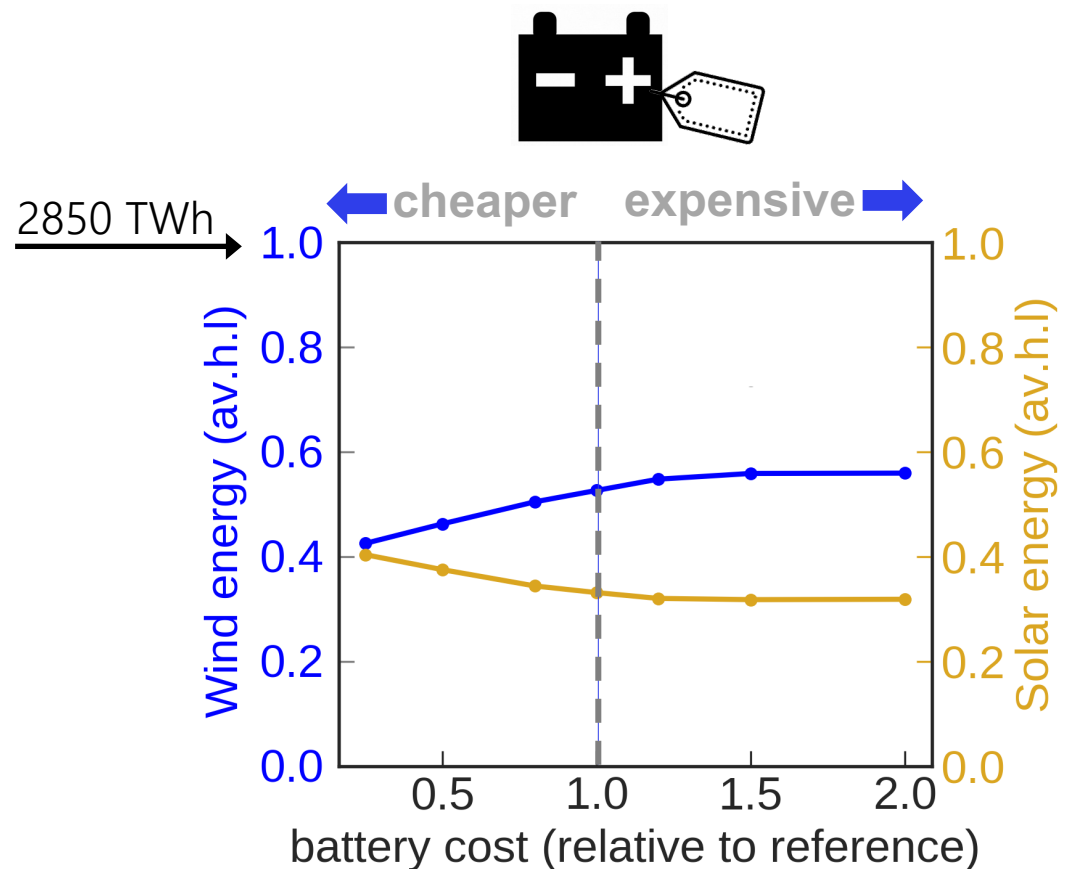
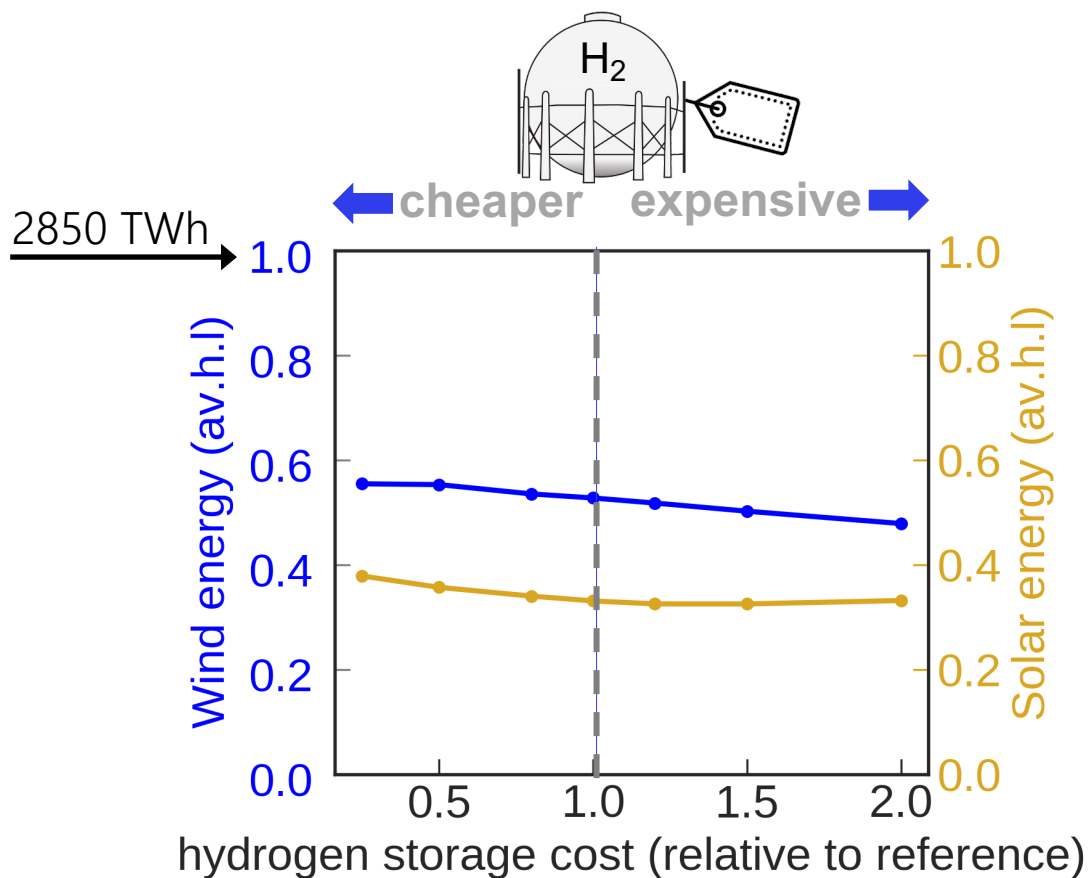
Brown et al., Energy 2018

DTU Sensitivity to wind and PV costs



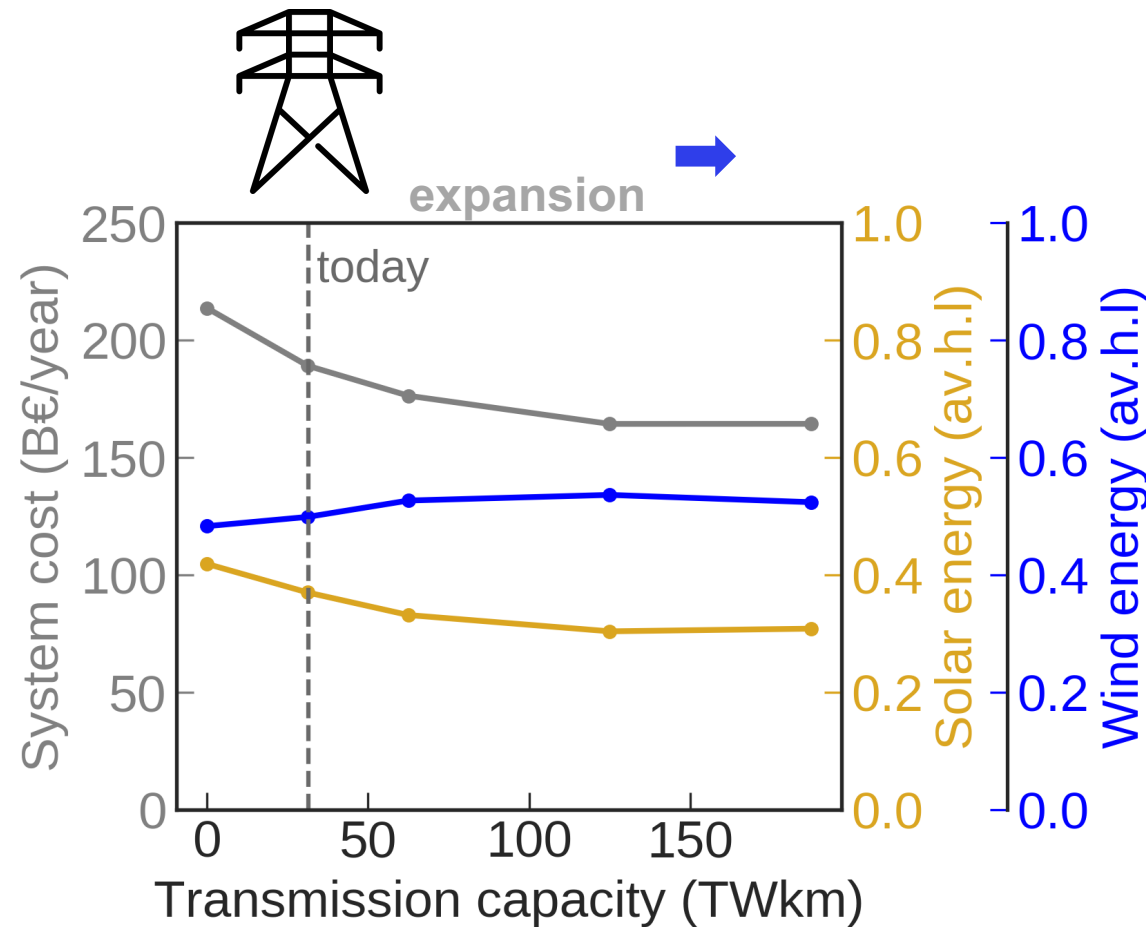
Even for very cheap PV, no 100% solar system is optimum as it would require large (expensive) battery capacity.

Sensitivity to battery and hydrogen storage costs



Cheap batteries increase optimal PV penetration.

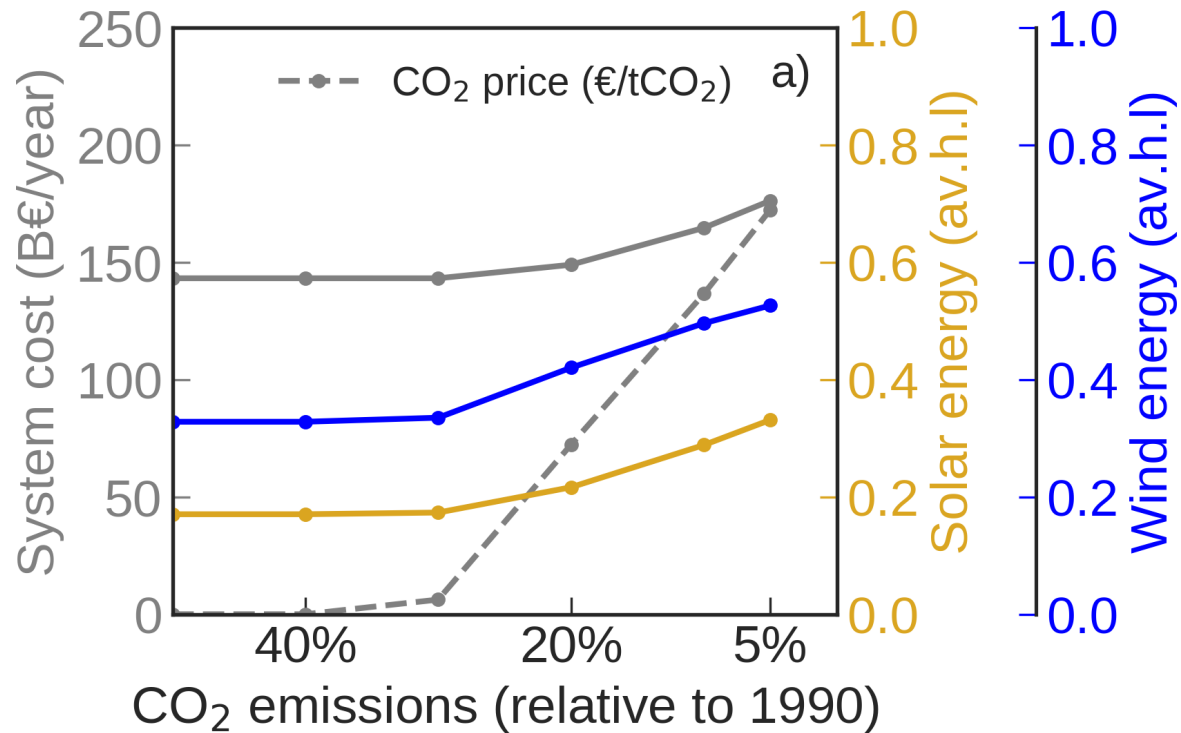
Impact of transmission capacity expansion



Increasing transmission capacity reduces PV optimal penetration.

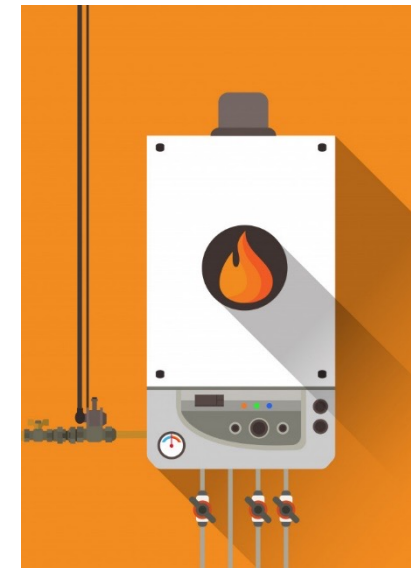
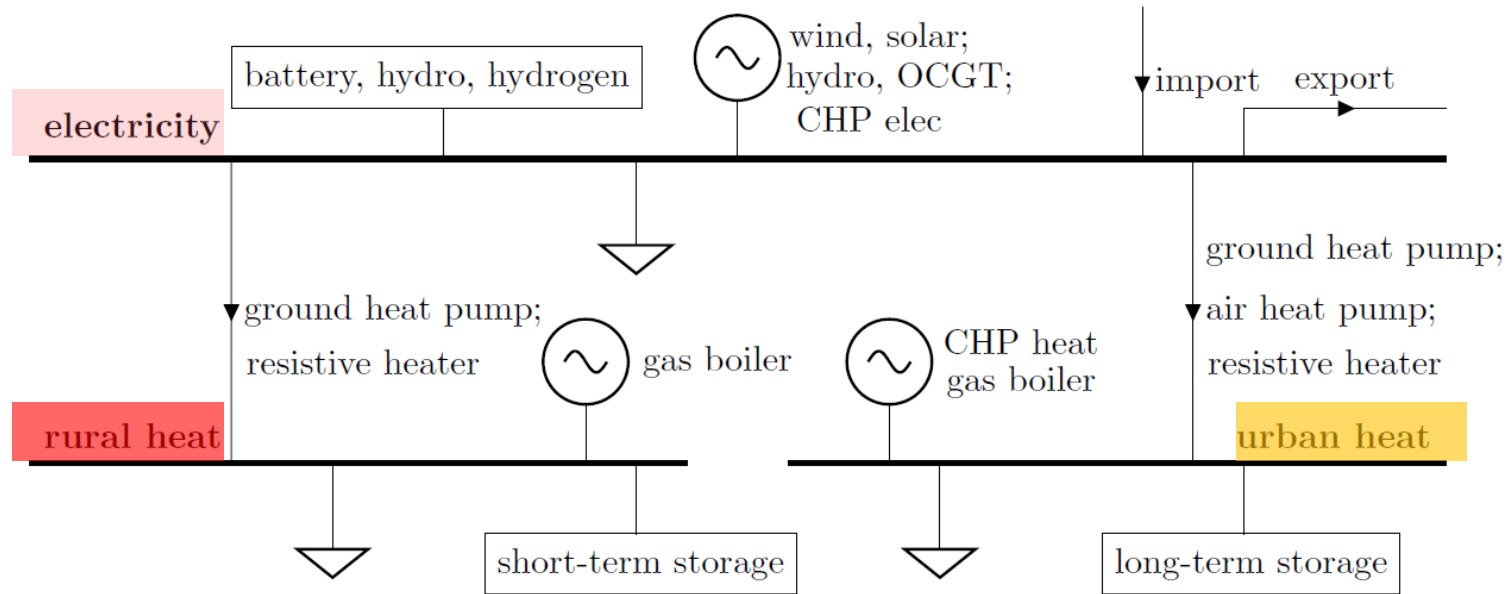
Increasing transmission capacity reduces system costs but most of the benefits are captured by the initial 25% grid expansion.

Sensitivity to CO₂ emissions allowance

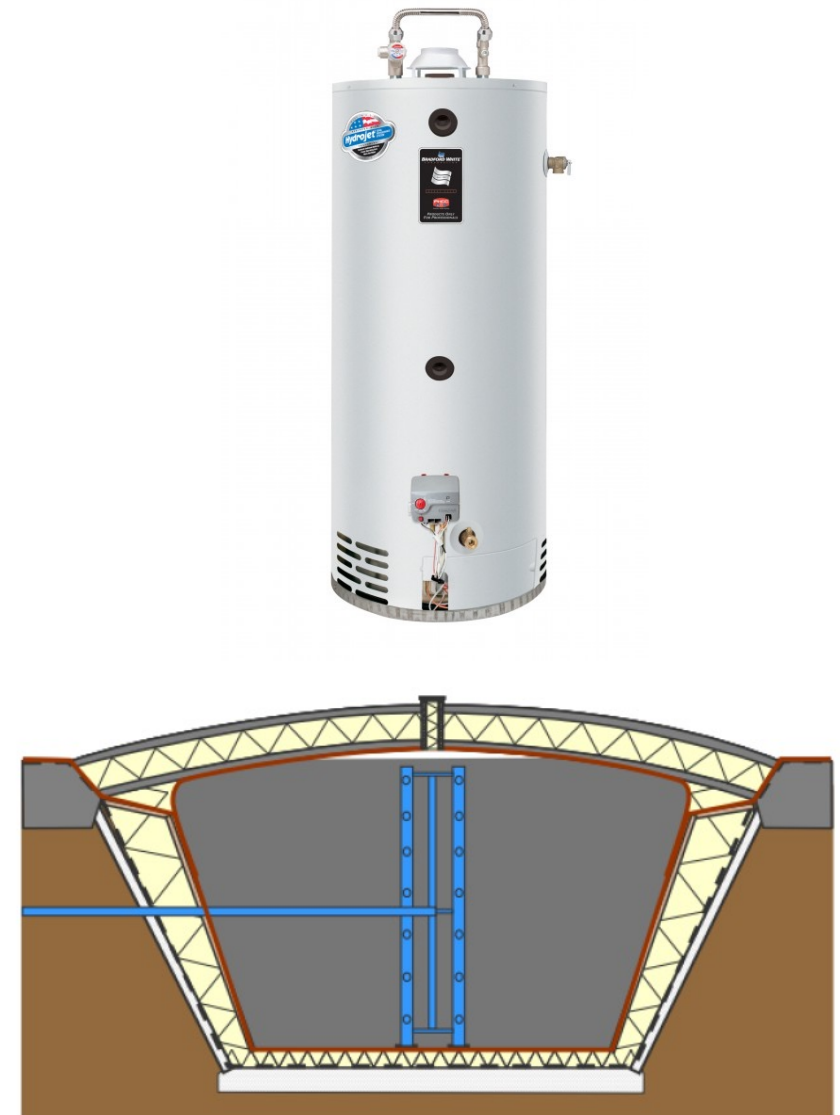
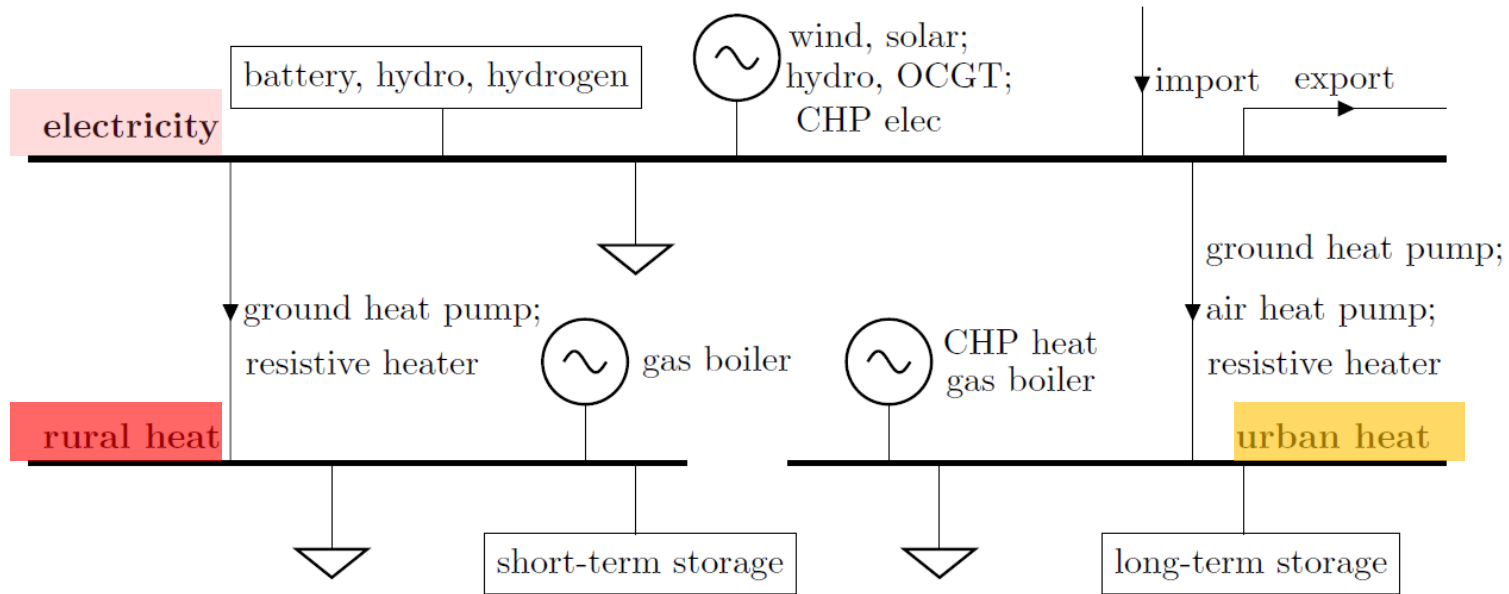


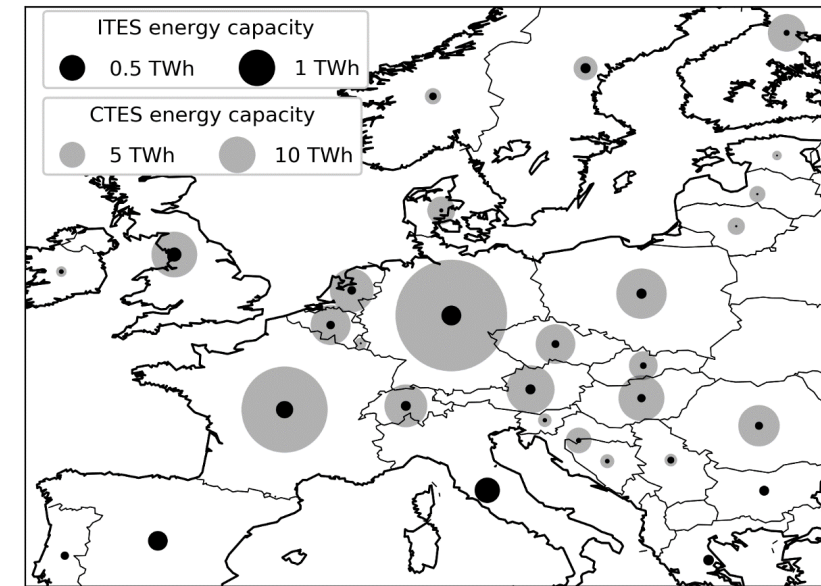
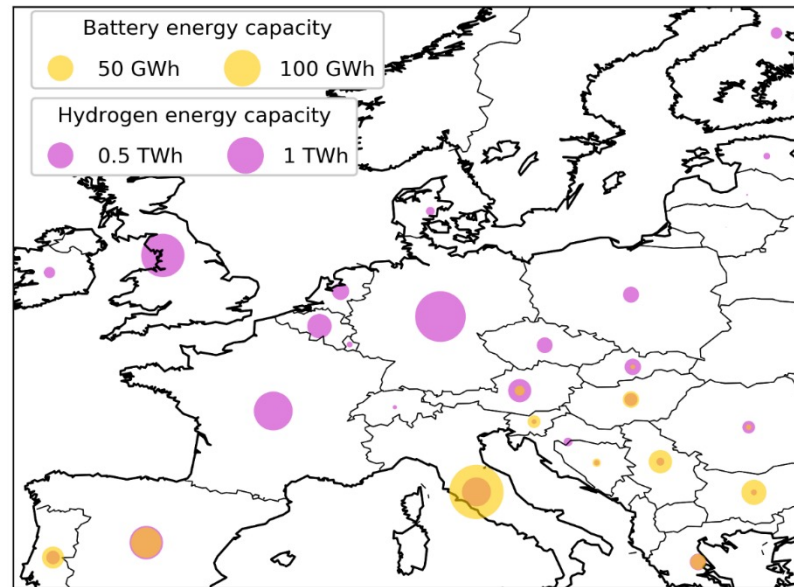
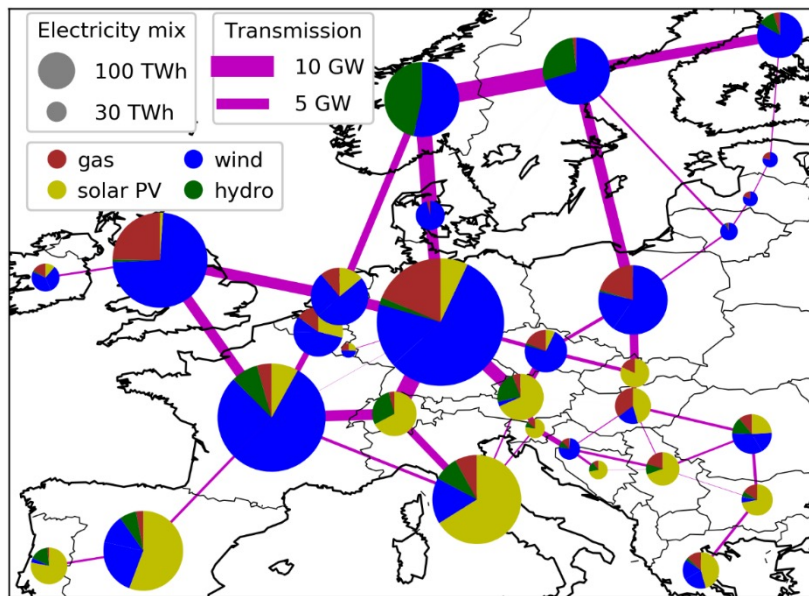
When CO₂ emissions are restricted, higher solar and wind capacities are needed. Higher CO₂ prices are also necessary.

Sector-coupled network model

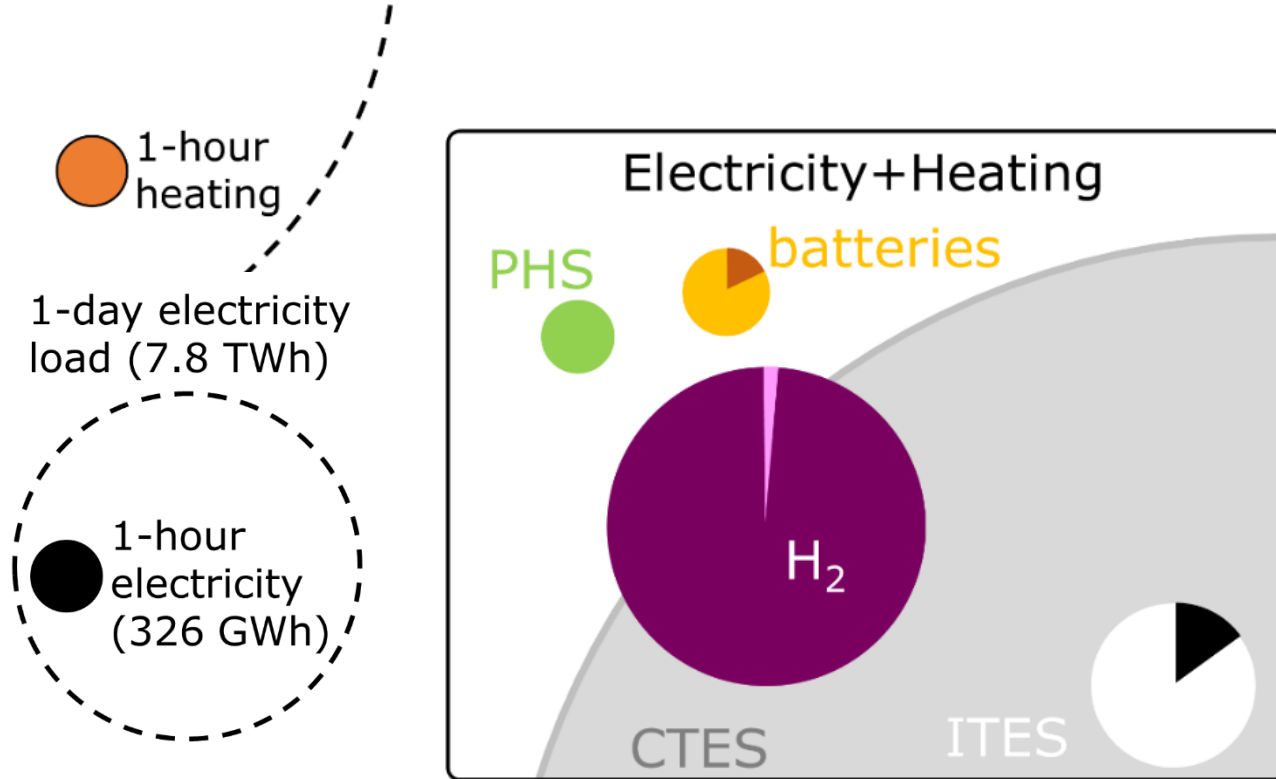


Sector-coupled network model



Electricity + heating scenario (5% CO₂ emissions, relative to 1990)

How much storage capacity is needed?



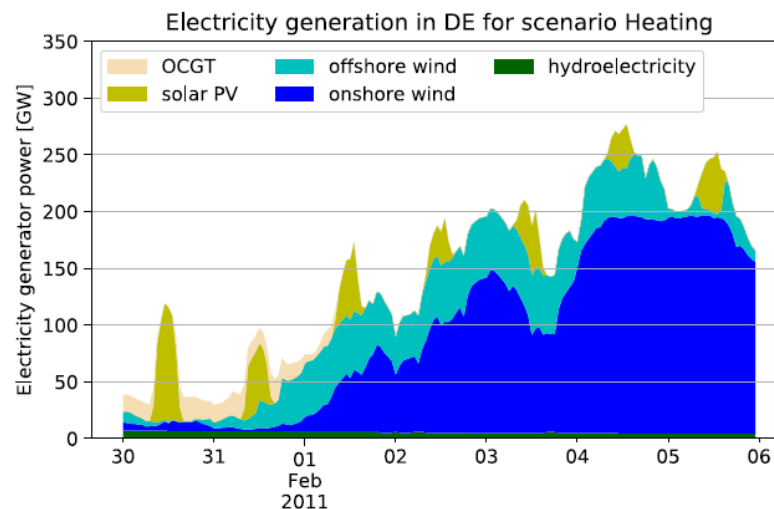
Individual thermal energy storage (ITES) sized to store heat demand for a few hours

Centralised thermal energy storage (CTES) sized to store heating demand for months

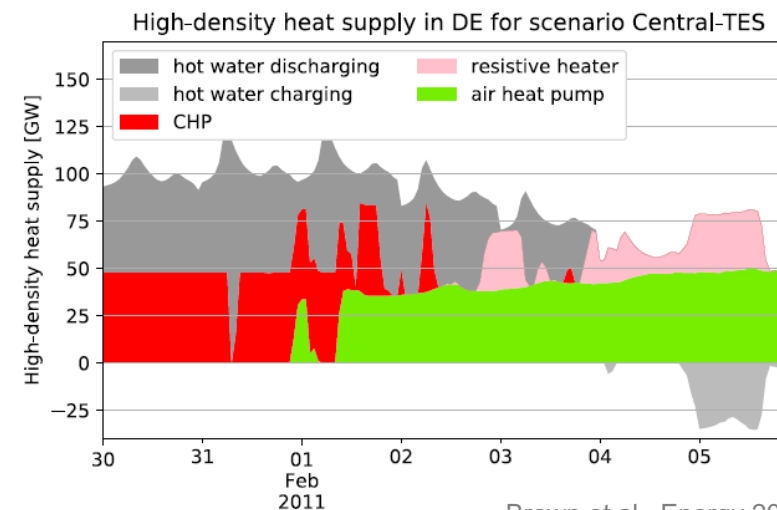
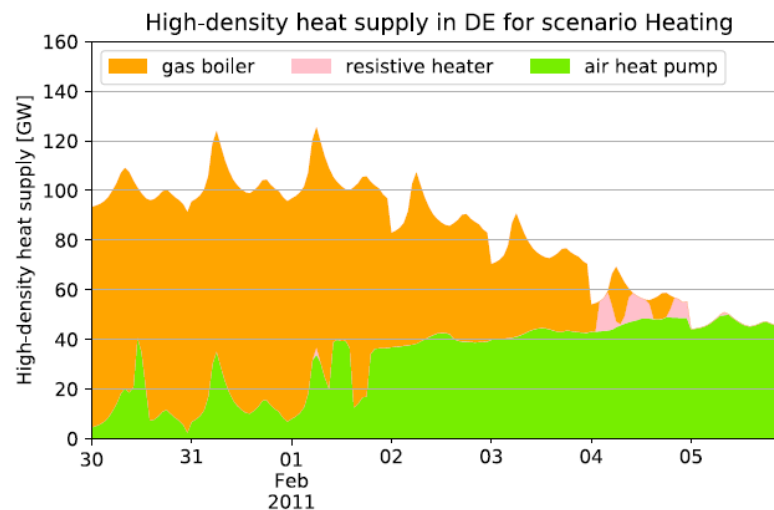
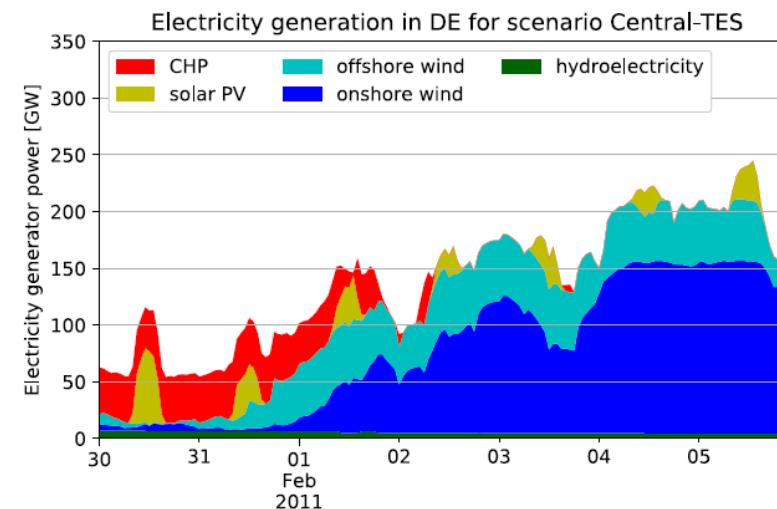
Victoria et al., Energy Conversion and Management (2019)

Centralised thermal energy storage (hot water tanks) supplies heat during cold spell (cold week with low solar and wind generation) avoiding the need of gas.

Without central TES

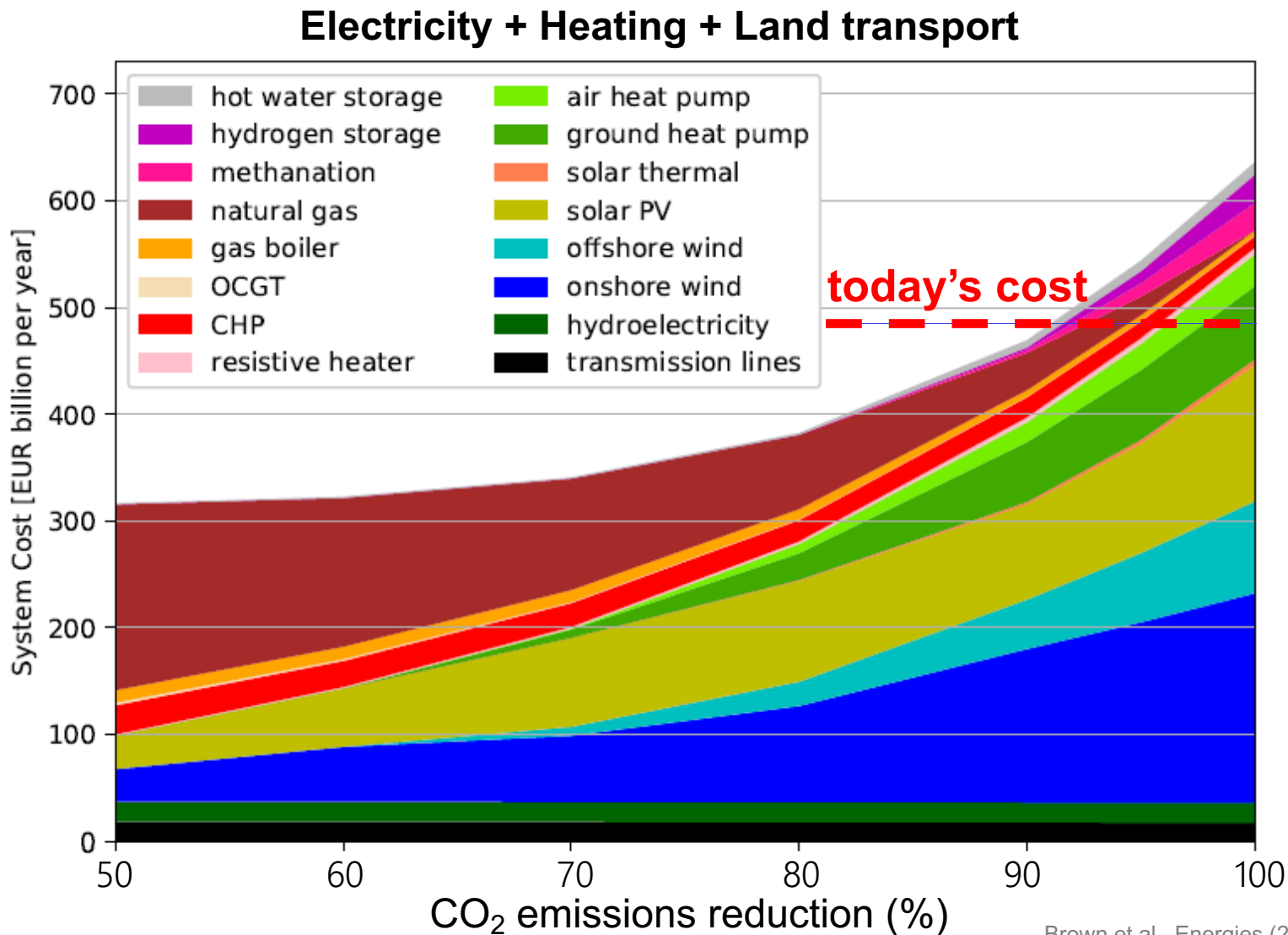


With central TES



Brown et al., Energy 2018

DTU Impact of CO₂ reduction target



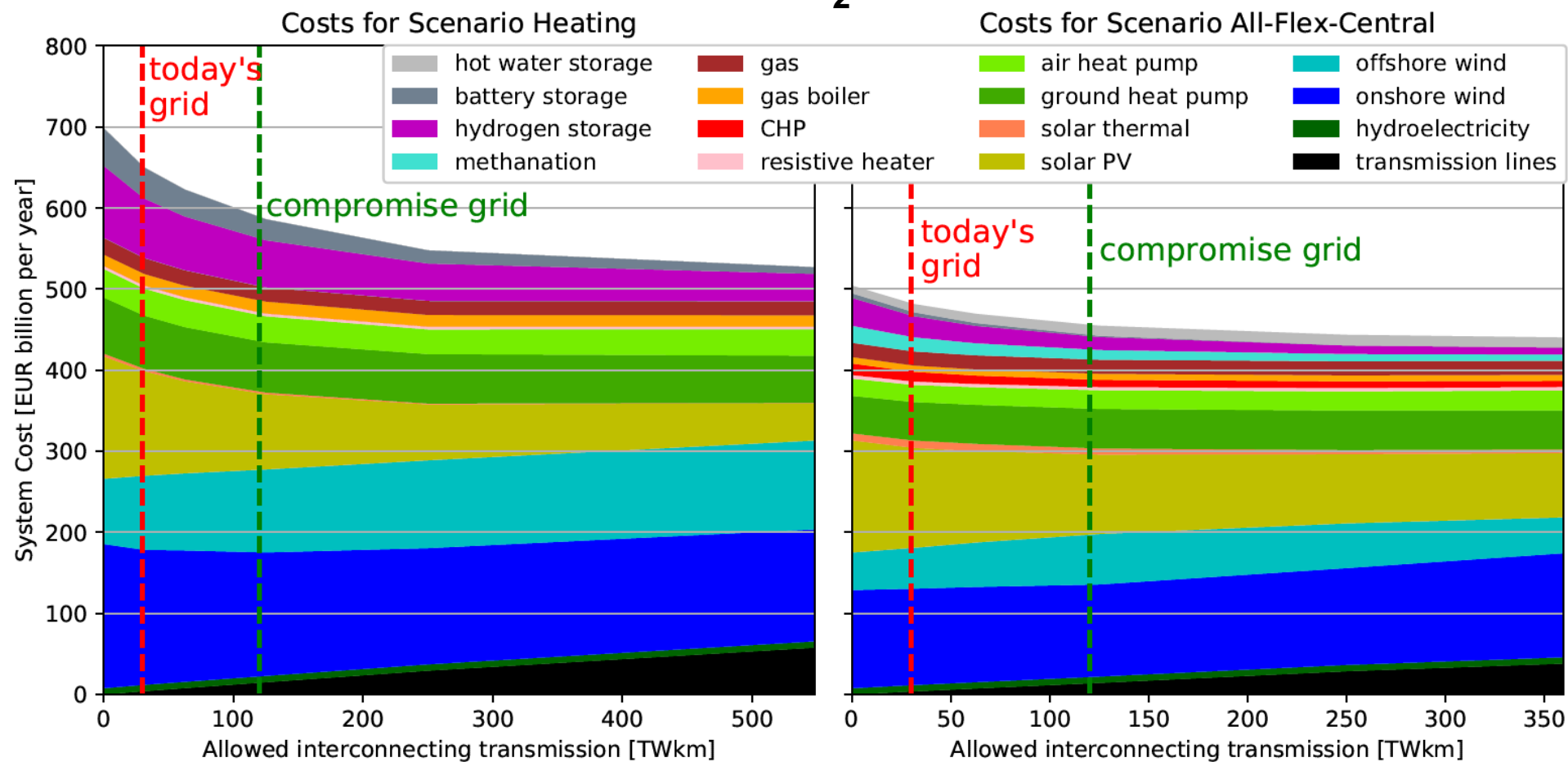
As CO₂ emissions are restricted the system becomes more expensive ...

... but not linearly, the last 20% is the hardest!

Brown et al., Energies (2019)

Impact of transmission grid expansion gets reduced

5% CO₂ limit



Brown et al., Energy 2018

Modelling land transport demand and flexibility

Challenge: Full electrification of land transport demand will add around 1000 TWh/a to current electricity demand in Europe (3000 TWh/a)

Opportunities: Efficiency increase, EV batteries bring large flexibility to the system, strong synergy with rooftop solar PV

Transportation

Internal-combustion engine (30% eff.)



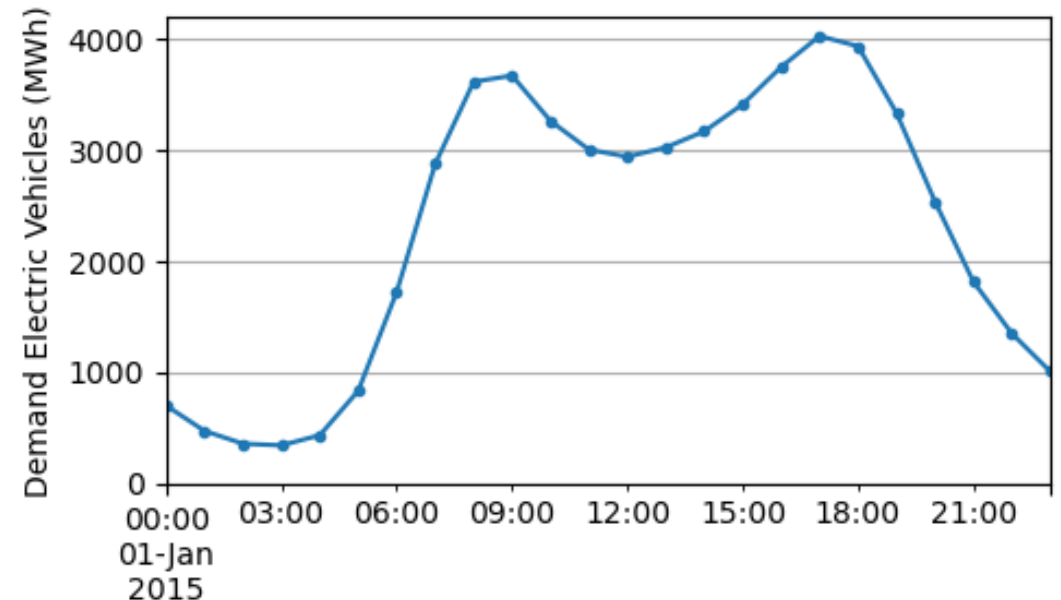
Electric vehicle



How can we model land transport demand?

Assume we want to estimate hourly-resolved demand for a fleet of Electric Vehicles (EVs).

1. Annual demand data for current cars fleet exists and can be transformed into electrified demand taking into account the different efficiency of Internal Combustion Engine (ICE) cars and EVs
2. Hourly-resolved data on cars usage can be used to create time series, e.g. assuming that EVs are charged right after they are used
3. We can model all the cars in a country/region lumped assuming average values (e.g. for every home-charger a capacity of 11 kW and 90% efficiency).



This represents unflexible demand from EVs.

How can we model land transport flexibility?

- **Smart-charging or Demand-Side Management (DSM)**

An EV battery can be added so that it is charged when is optimal for the system.

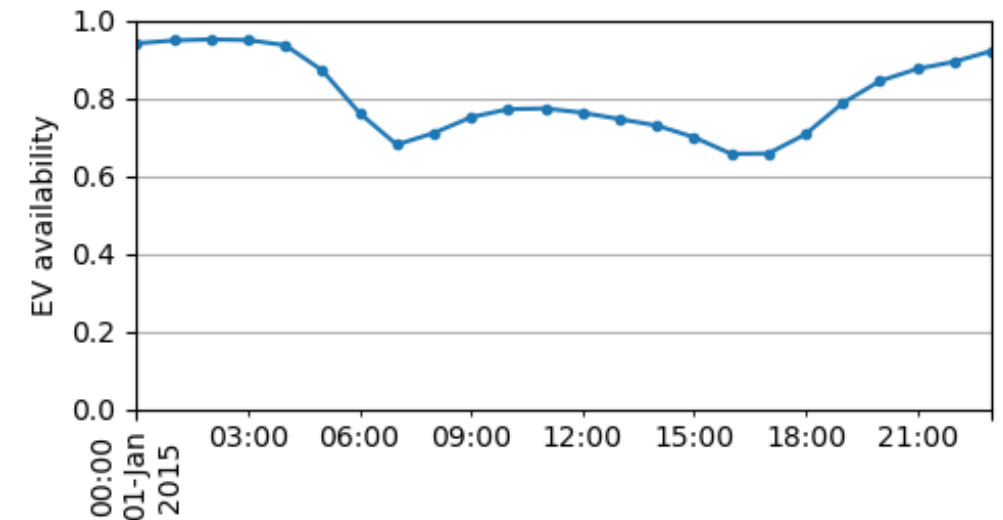
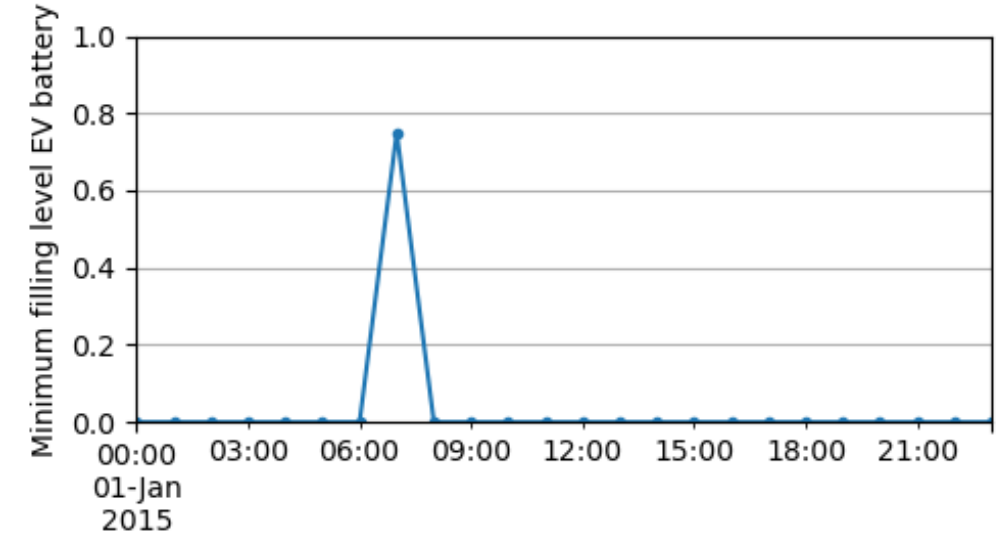
We can model all the cars in a country/region lumped assuming for every EV an average capacity (e.g. 50 kWh)

To make the operation of the EV batteries more realistic, we can also impose that their state of charge $> 75\%$ at 7 a.m. every day

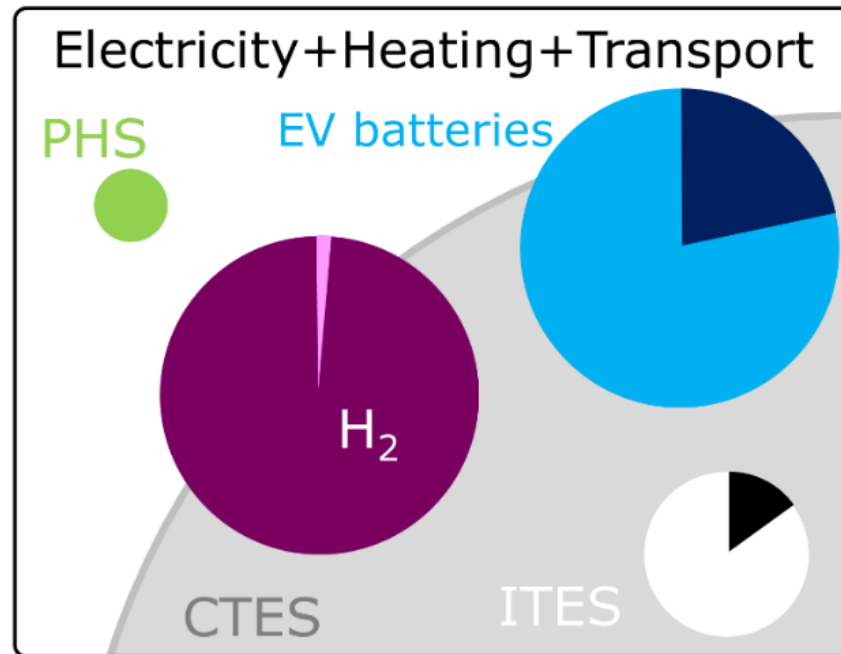
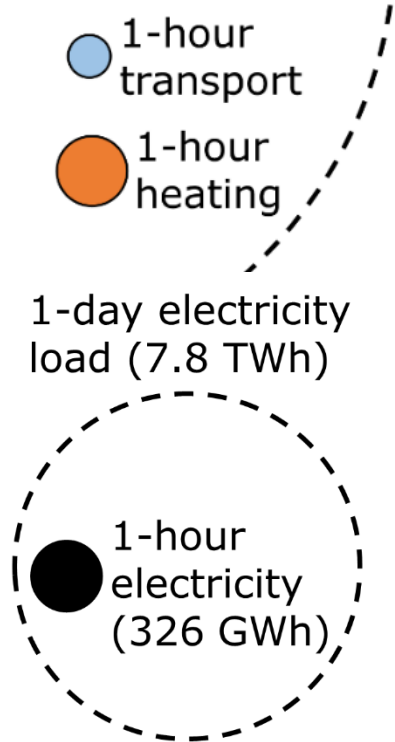
- **Vehicle-to-Grid (V2G)**

We can also include the hypothesis that a percentage of cars allow vehicle-to-grid services.

We can calculate some availability profile (based on when cars are not been used).



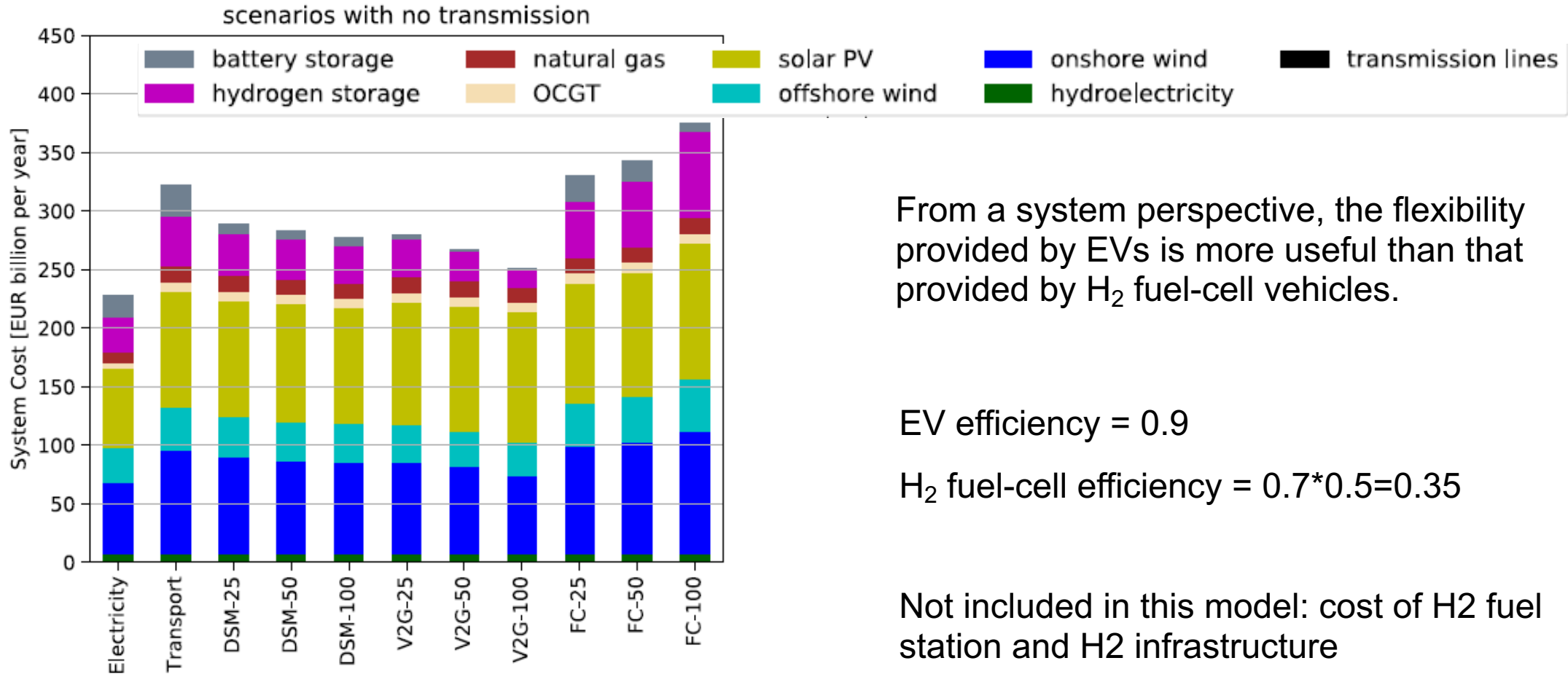
How much storage capacity is needed?



EV battery capacity could be enough to remove the need for static batteries

Victoria et al., Energy Conversion and Management (2019)

Electric Vehicles vs Hydrogen Vehicles



From a system perspective, the flexibility provided by EVs is more useful than that provided by H₂ fuel-cell vehicles.

EV efficiency = 0.9

H₂ fuel-cell efficiency = $0.7 \times 0.5 = 0.35$

Not included in this model: cost of H₂ fuel station and H₂ infrastructure

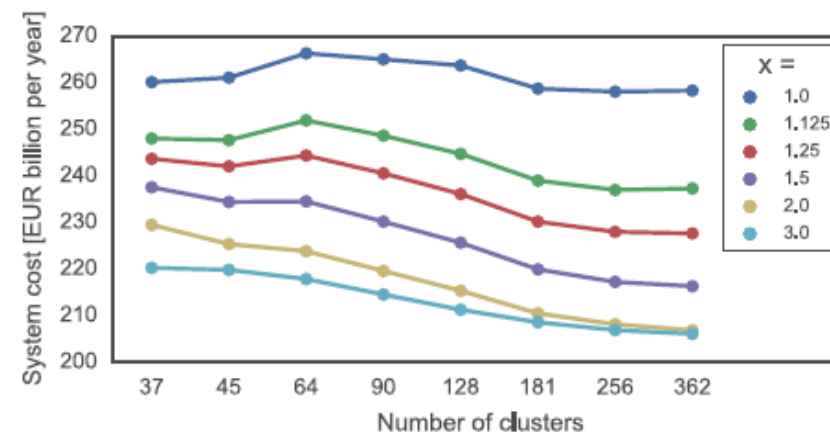
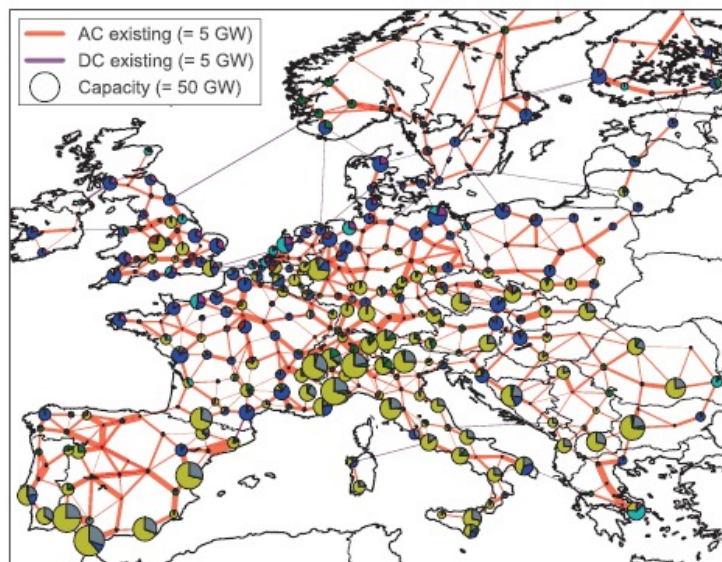
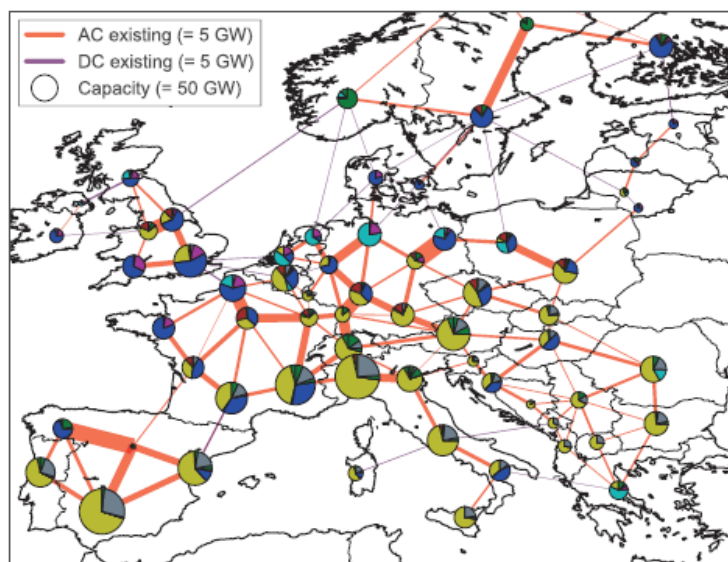
Brown et al., Energy 2018

Which temporal and spatial resolution is good enough?

Which spatial resolution is good enough?

When increasing the number of nodes, the system cost remains roughly constant due to the counterbalancing of two effects:

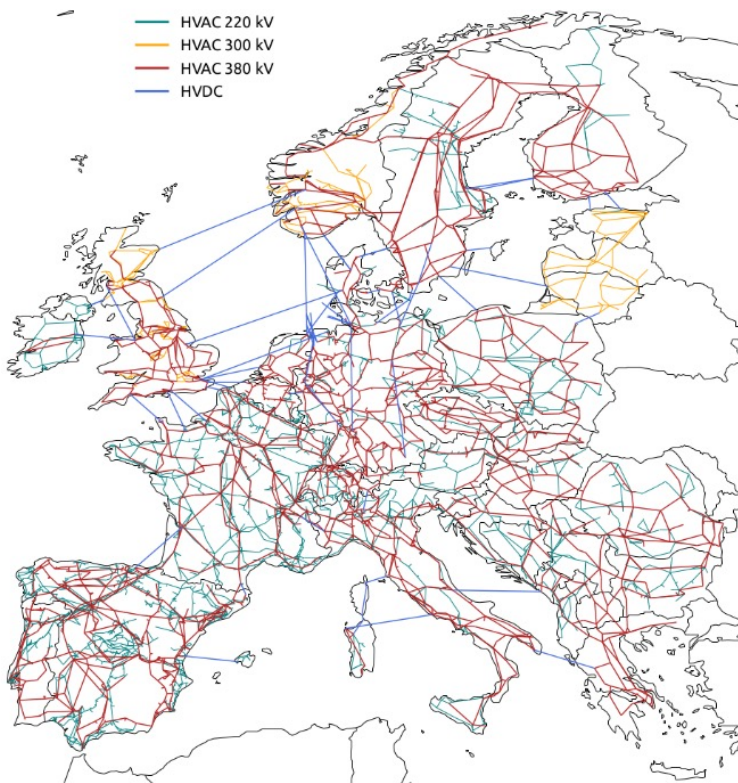
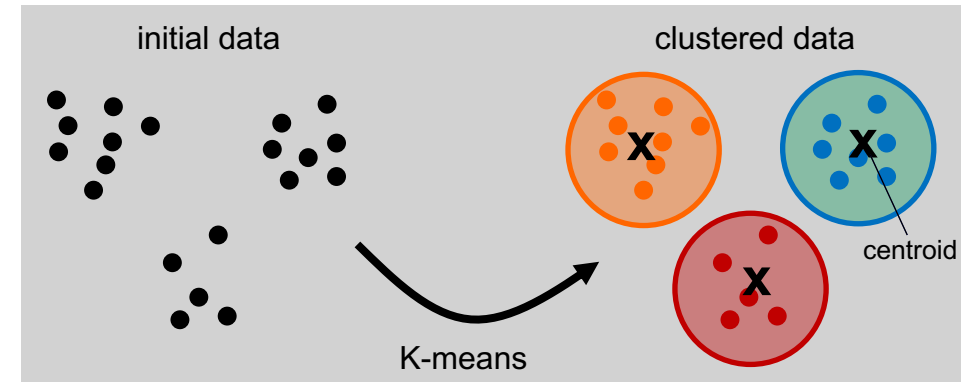
- (a) sites with high capacity factors for wind and solar are available for a more finely resolved network,
- (b) but the emergence of bottlenecks inside countries prevents the use wind energy generated at exterior nodes



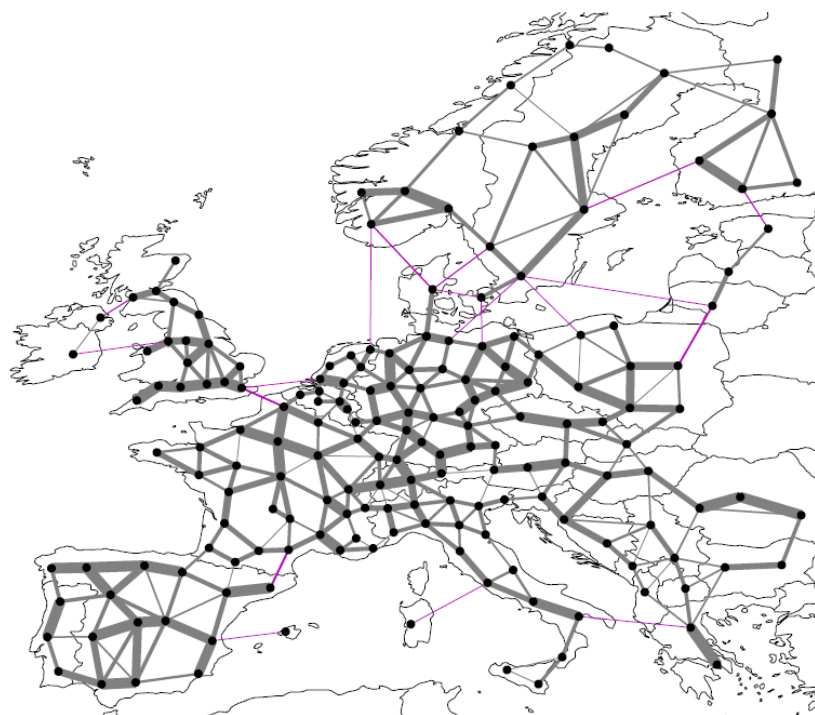
Hörsch and Brown, IEEE (2017)

How can we cluster power networks?

We cluster the network using k-means algorithms.

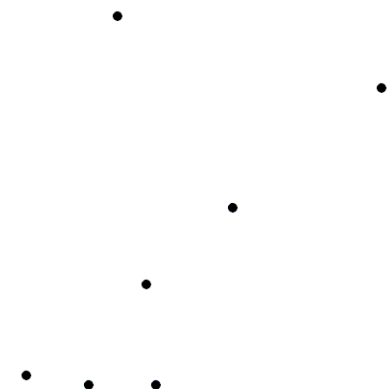


Existing HVAC and HVDC network in Europe



Network clustered to 181 nodes

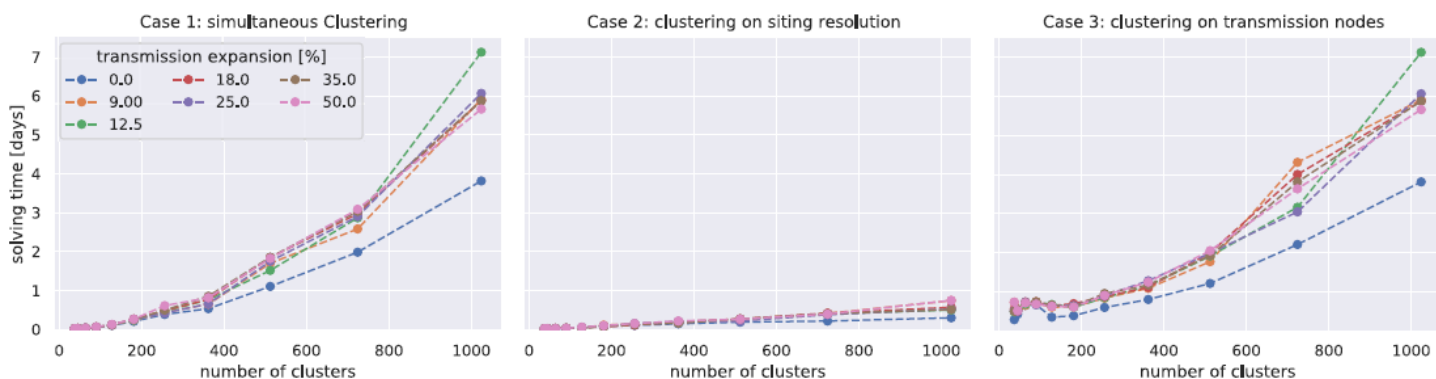
The area is split using Voronoi cells
(each point assigned to the closest node)



Source: [wikipedia](https://en.wikipedia.org/wiki/Voronoi_diagram)

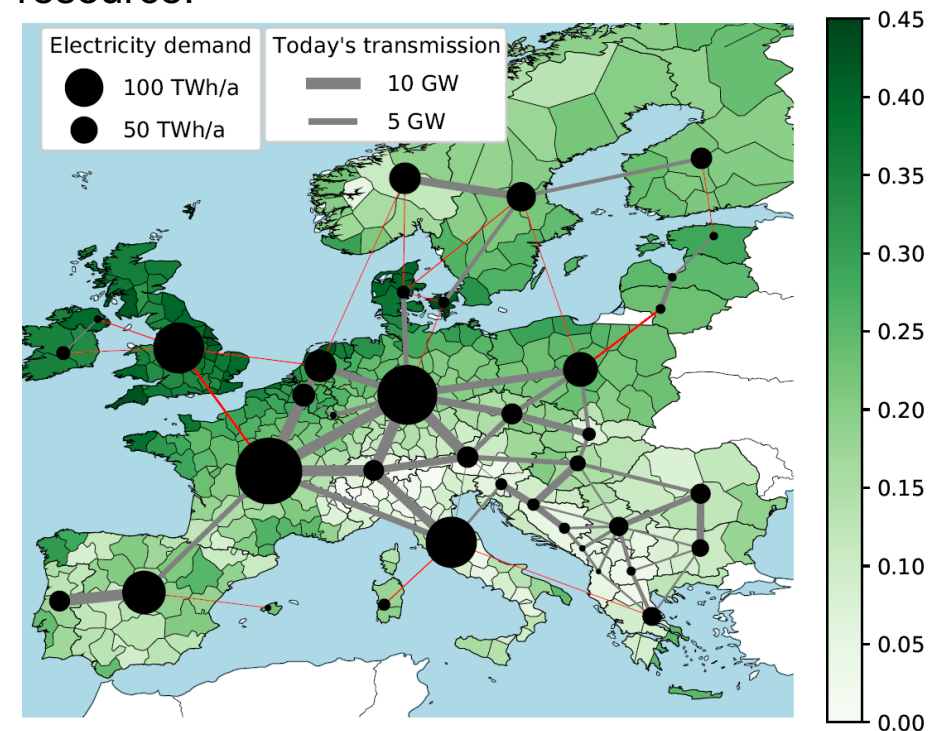
Which spatial resolution is good enough?

Attaining high resolution on transmission nodes increases solving time, but increasing (only) resolution on sites for renewable has negligible impact.



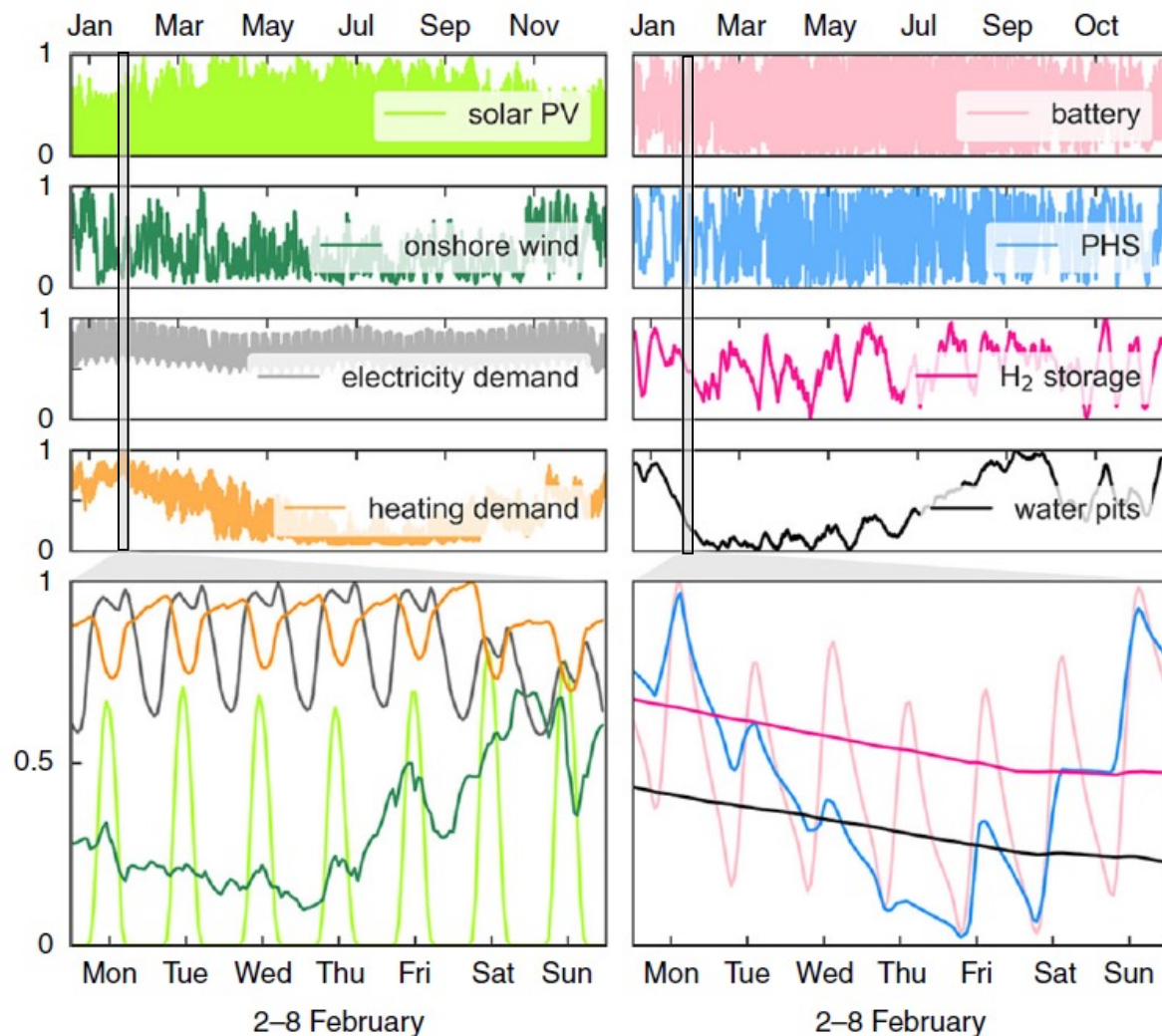
Frysztacki et al., Applied Energy (2021)

In some of our recent works, we keep a 37-node network and 370 nodes to resolve solar and wind resource.



Victoria et al., Joule (2022)

Which time resolution is good enough?



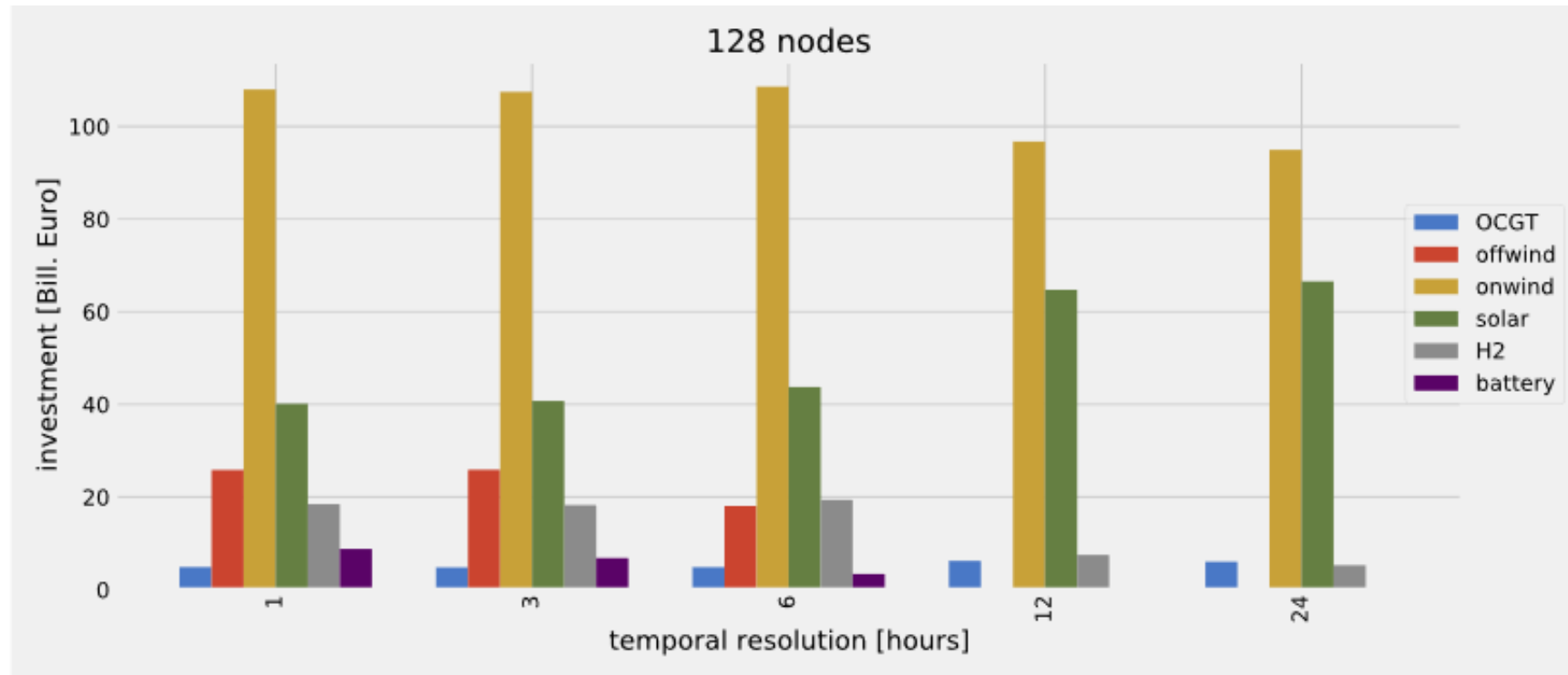
Victoria et al., Nature Commun. (2020)

We need **uninterrupted hourly time stepping for a full year** to capture the main fluctuations:

- solar and wind power generation smoothed by the grid and storage
- the role of long-term storage
- system operation during dark doldrums (i.e., periods with low wind and solar generation)

Which time resolution is good enough?

Resolution coarser than 3 hours get a solar balance “for free” and does not need batteries

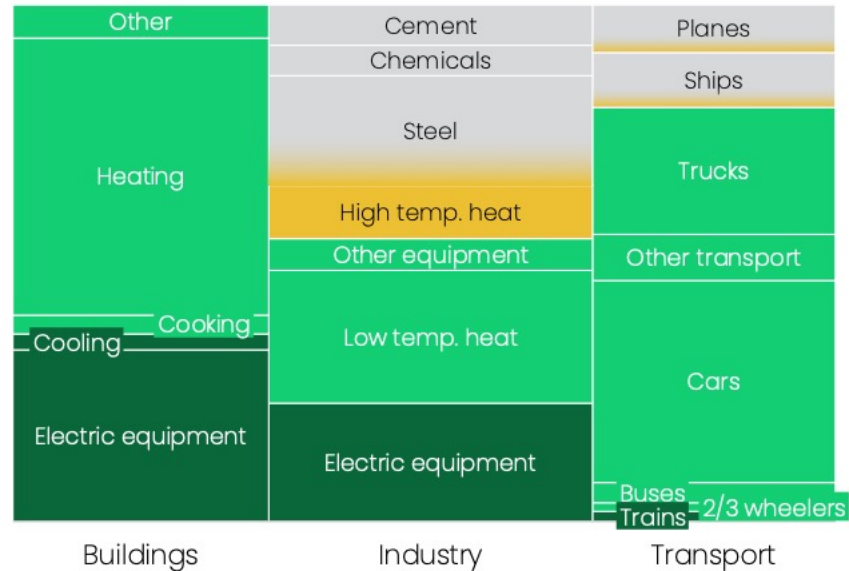


Schyska et al., Joule. (2021)

Preview: what if we can't electrify?

Share of final energy demand by subsector and electrification potential (%)

2025



■ Already (largely) electrified ■ Can be electrified technically
■ Can be electrified economically ■ Still under development

Source: [Ember: the electrotech revolution](#)





Problems for this lecture

Problems 11.1 (**Group 23**)

Problems 11.2 (**Group 24**)

DTU

